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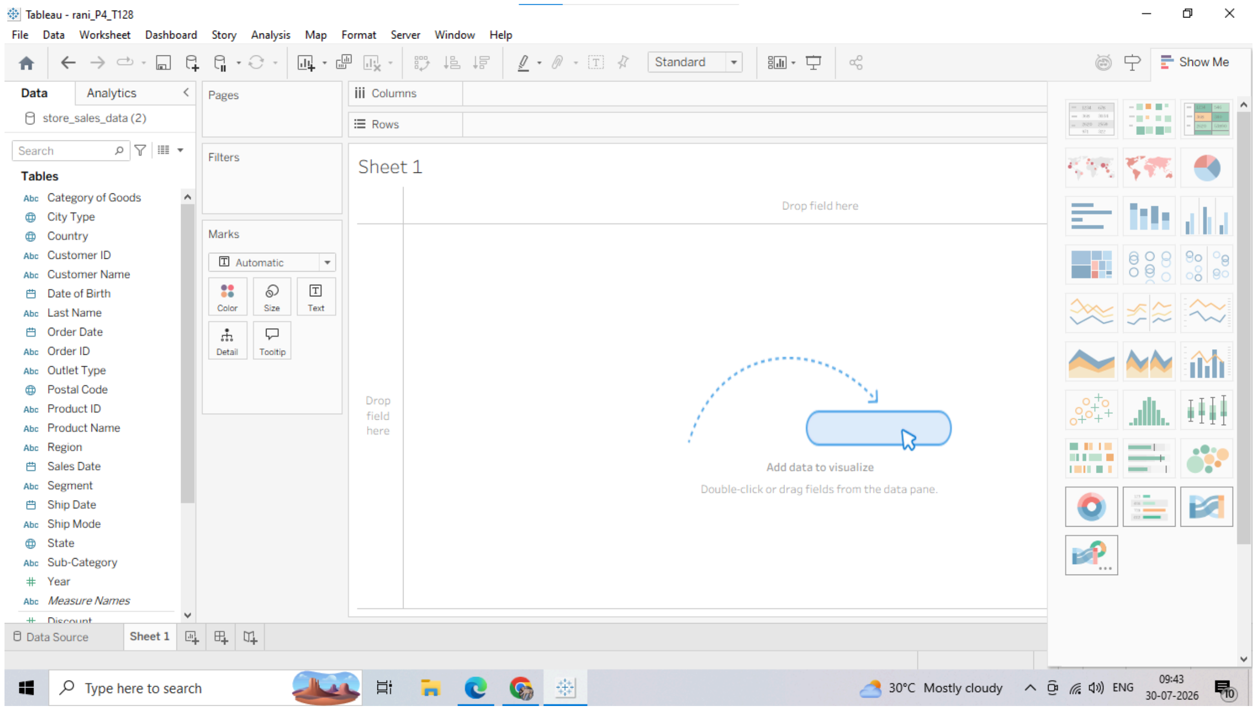
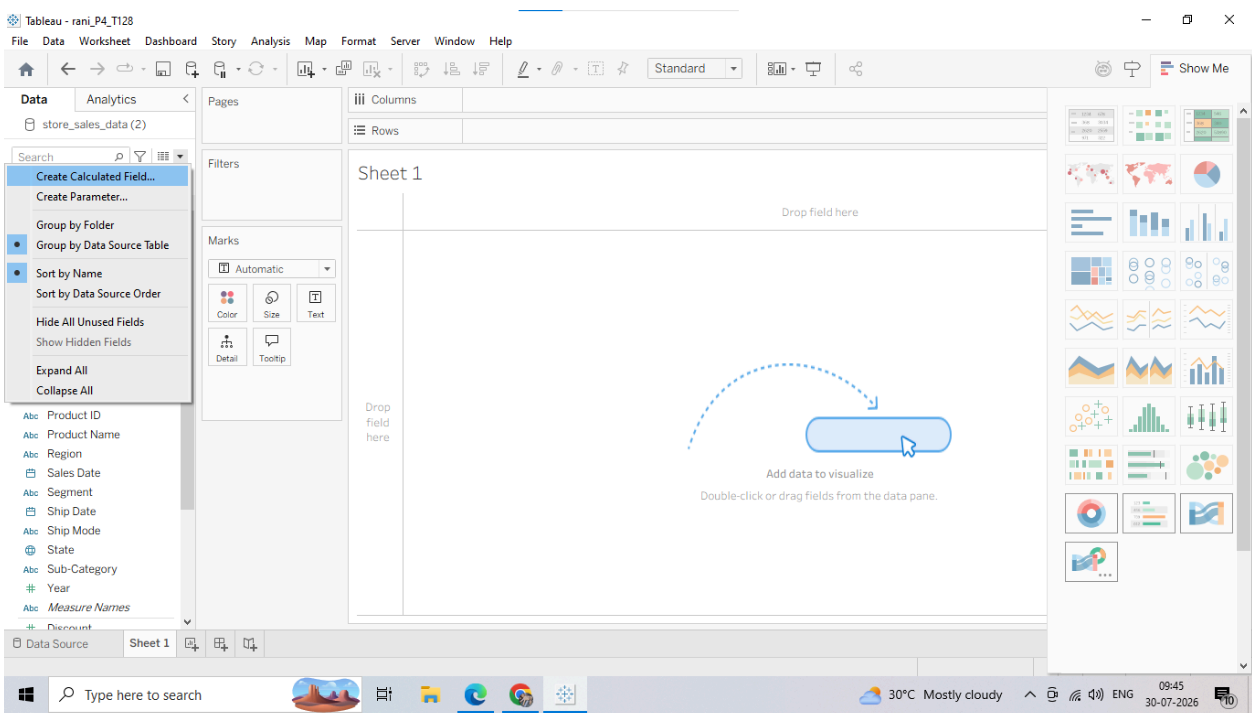
Click on Sheet 1 at the bottom-left corner to enter your workspace.



2. Building a Basic Custom Calculation

1. Look at your left-hand Data Pane. Right-click anywhere in the blank space of the pane (or click the small downward triangle next to the search bar) and select Create Calculated Field...



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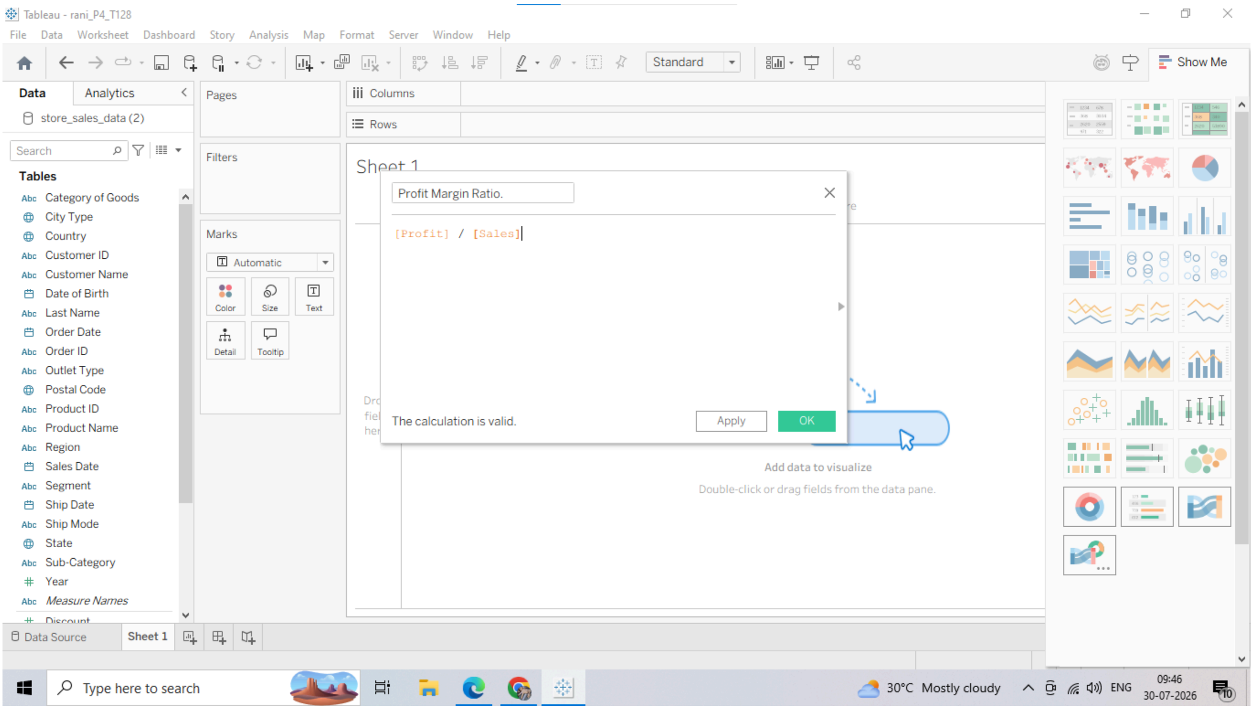
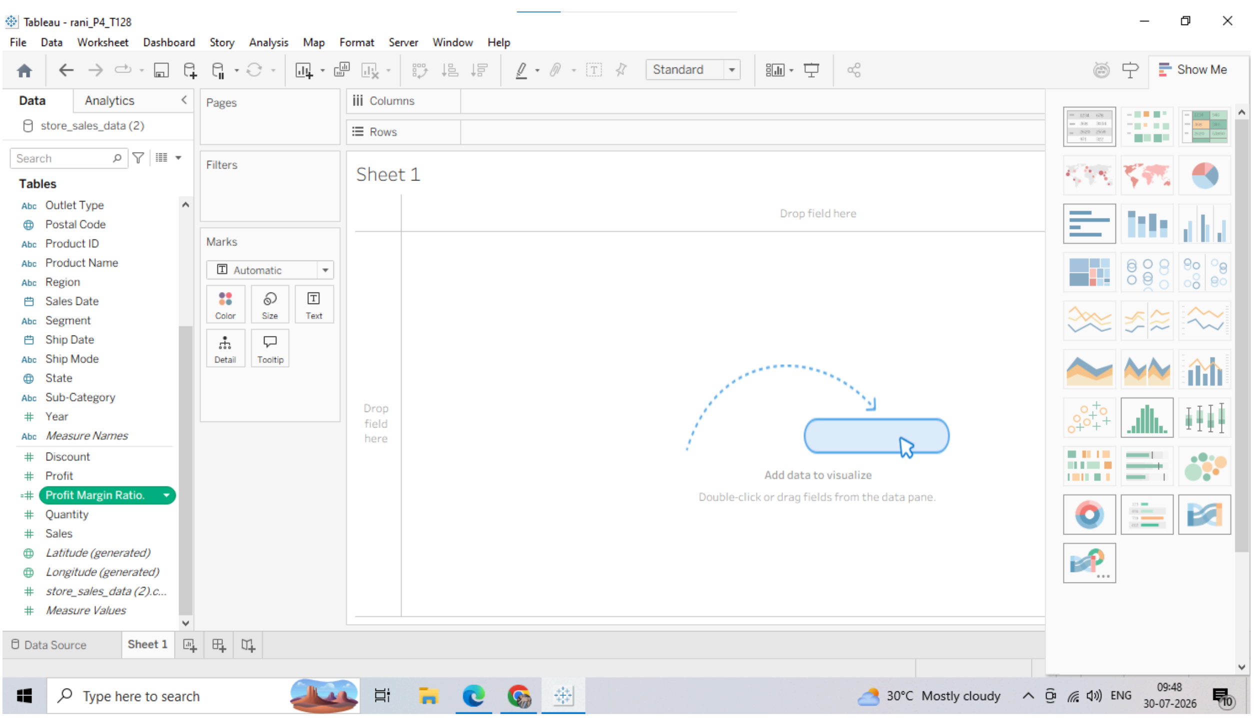
In the calculation editor window that pops up, clear any default text and name the field: Profit Margin Ratio.

In the main workspace box, enter the following basic mathematical formula:

[Profit] / [Sales]



Notice that your new custom field Profit Margin Ratio now appears in your Data Pane list with a small equal sign (=) next to its data type icon, showing it is a calculated field.



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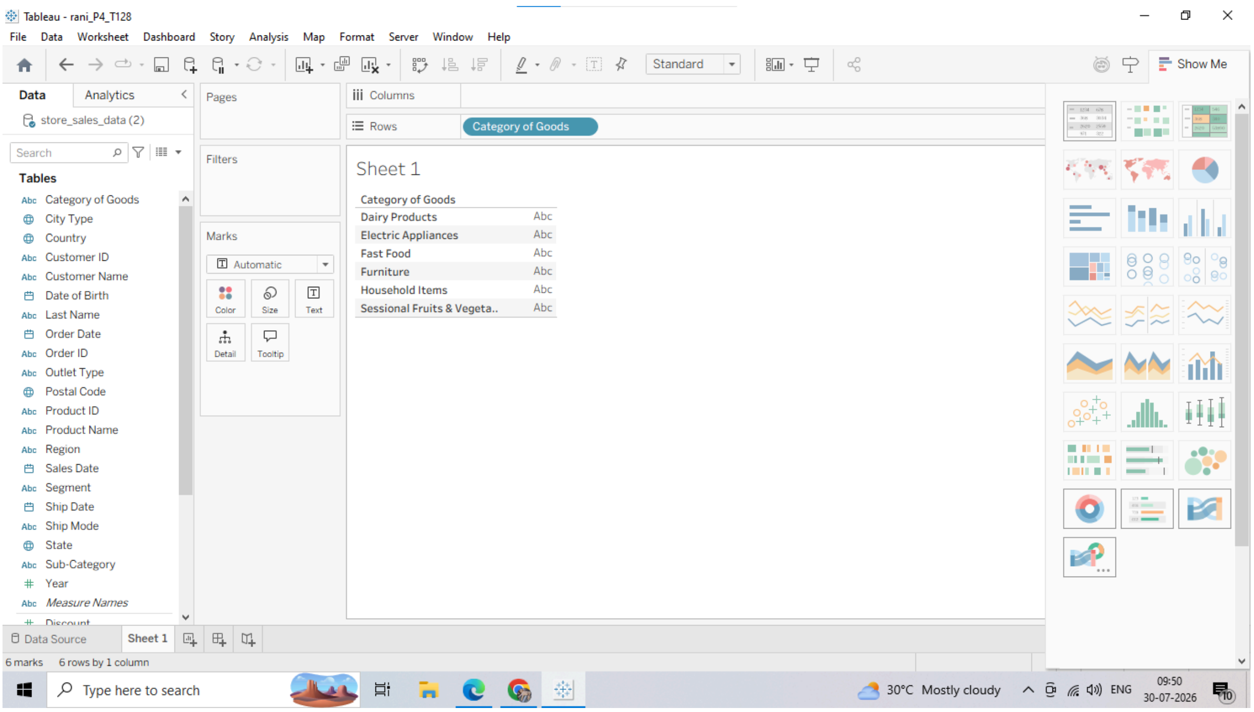
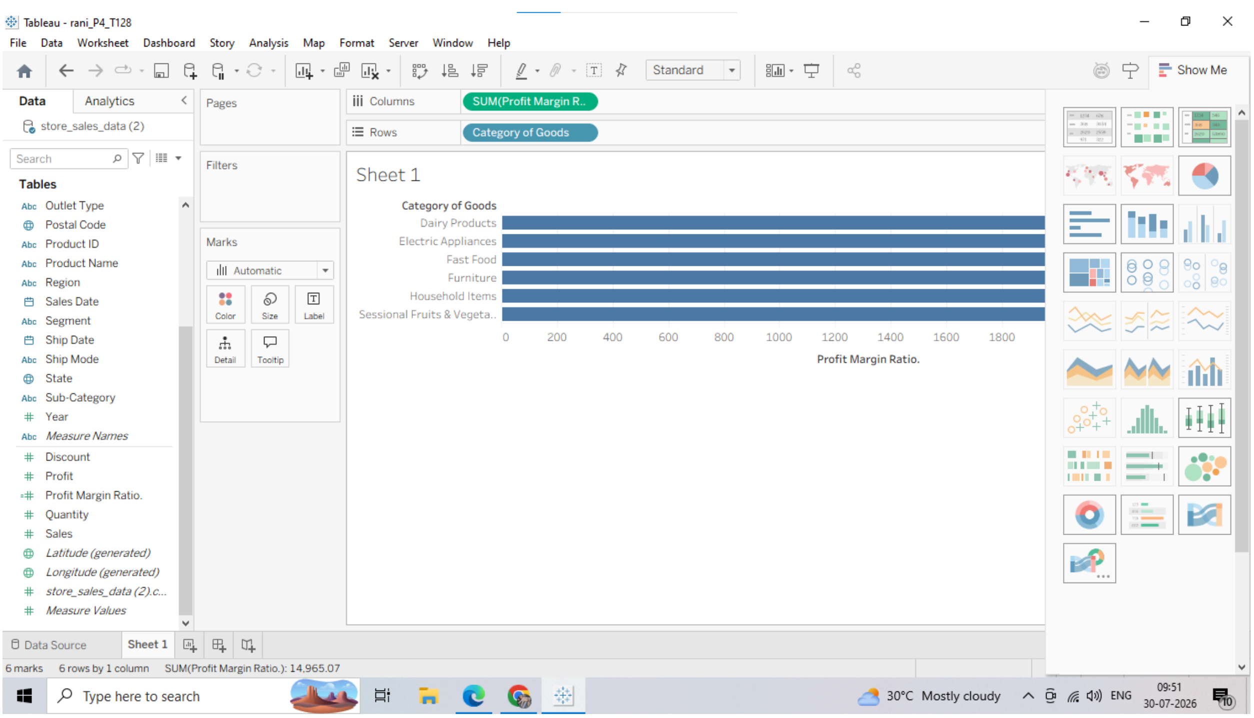
2. Putting the Field to Use

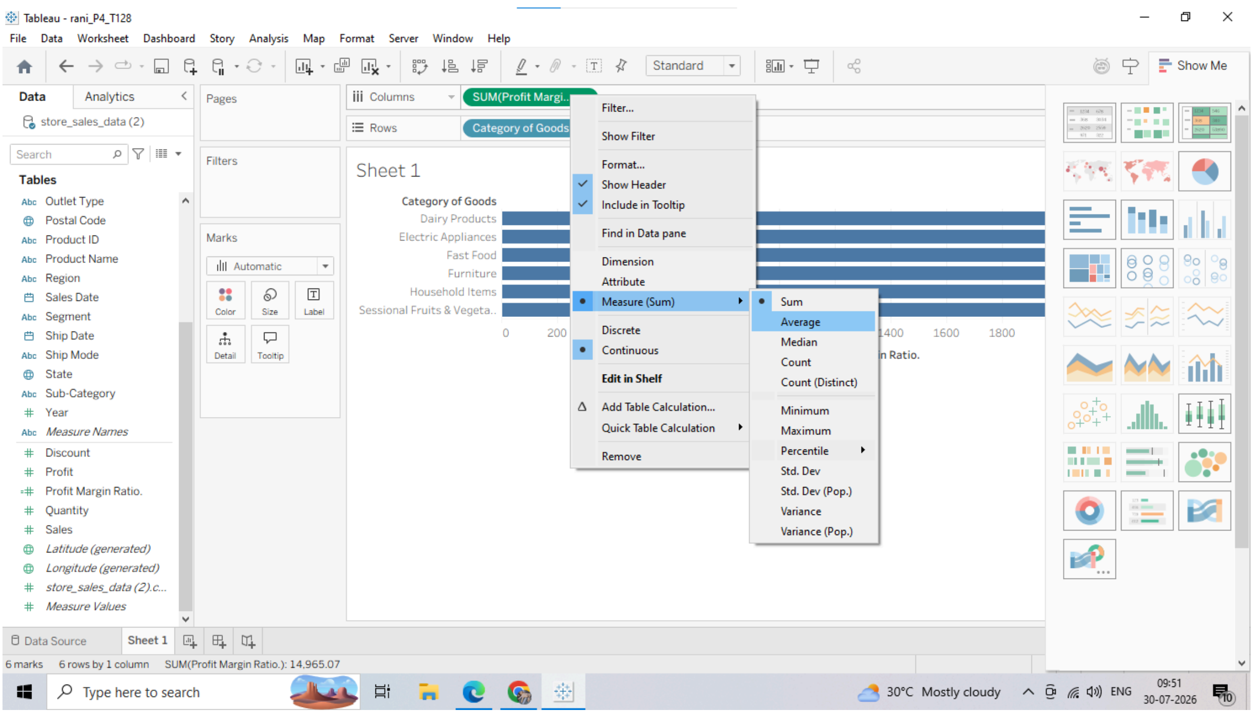
1. Drag the Category of Goods dimension from your Data Pane and drop it onto the Rows shelf.

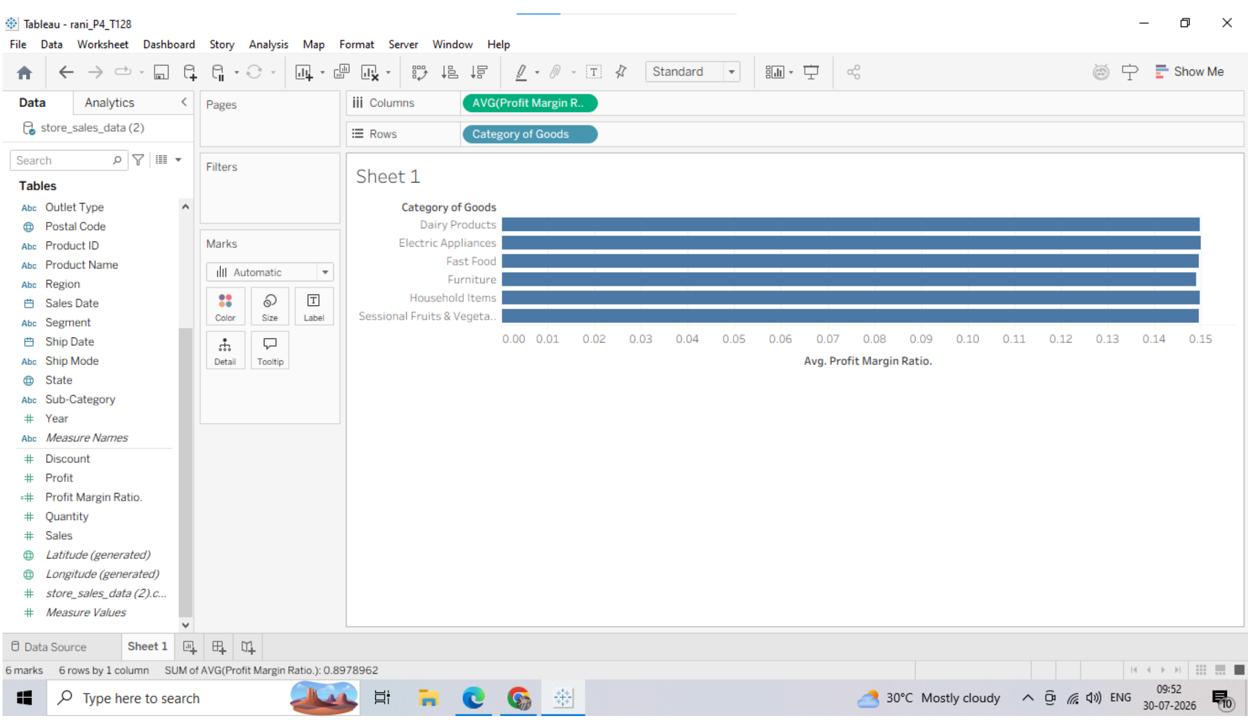


2. Drag your newly created Profit Margin Ratio calculated field and drop it onto the Columns shelf.



3. Right-click the Profit Margin Ratio pill now sitting on the Columns shelf, hover over Measure (Sum), and change it to Average to see the true margin distribution across product types.





Why we are doing it

A Basic Calculated Field lets you use standard arithmetic operators (+, -, *, /) to create brand new metrics that aren't available in the raw data files. We are calculating a profit margin ratio to evaluate efficiency rather than looking blindly at absolute volume numbers.

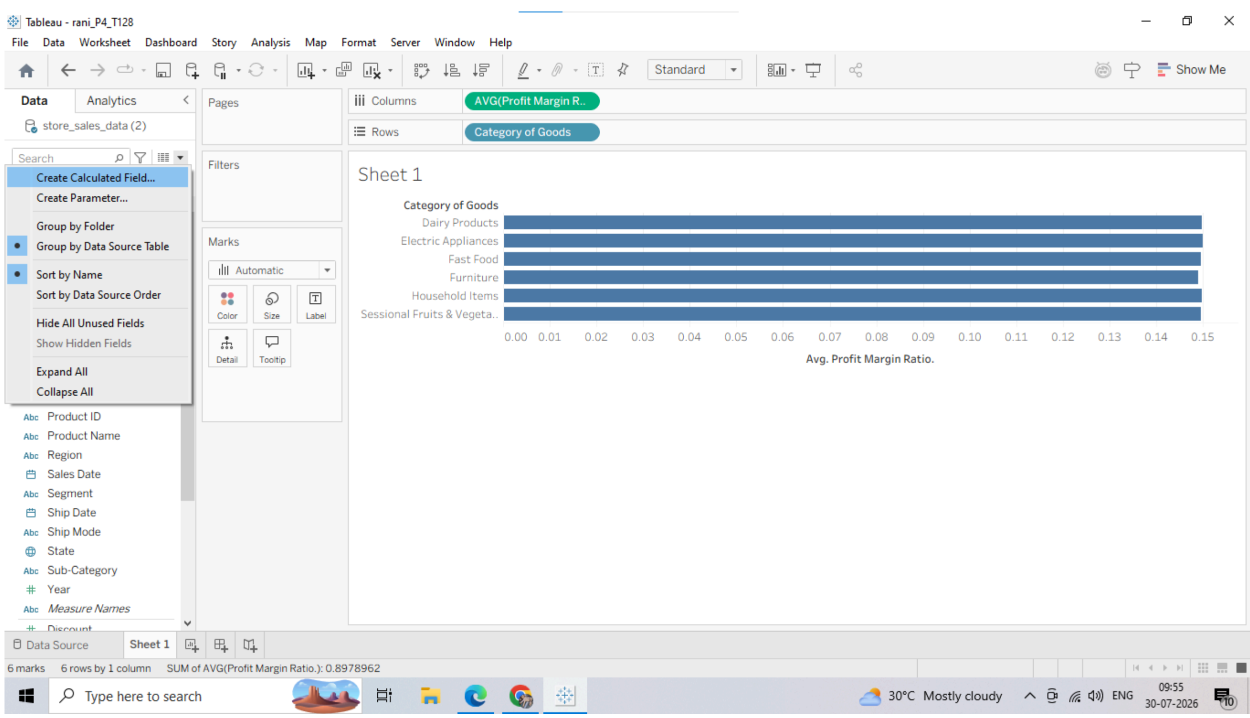
Usage

Use basic formulas whenever you need to calculate financial margins, combine text strings (like merging First Name and Last Name), or perform basic arithmetic scaling on individual rows before plotting them.

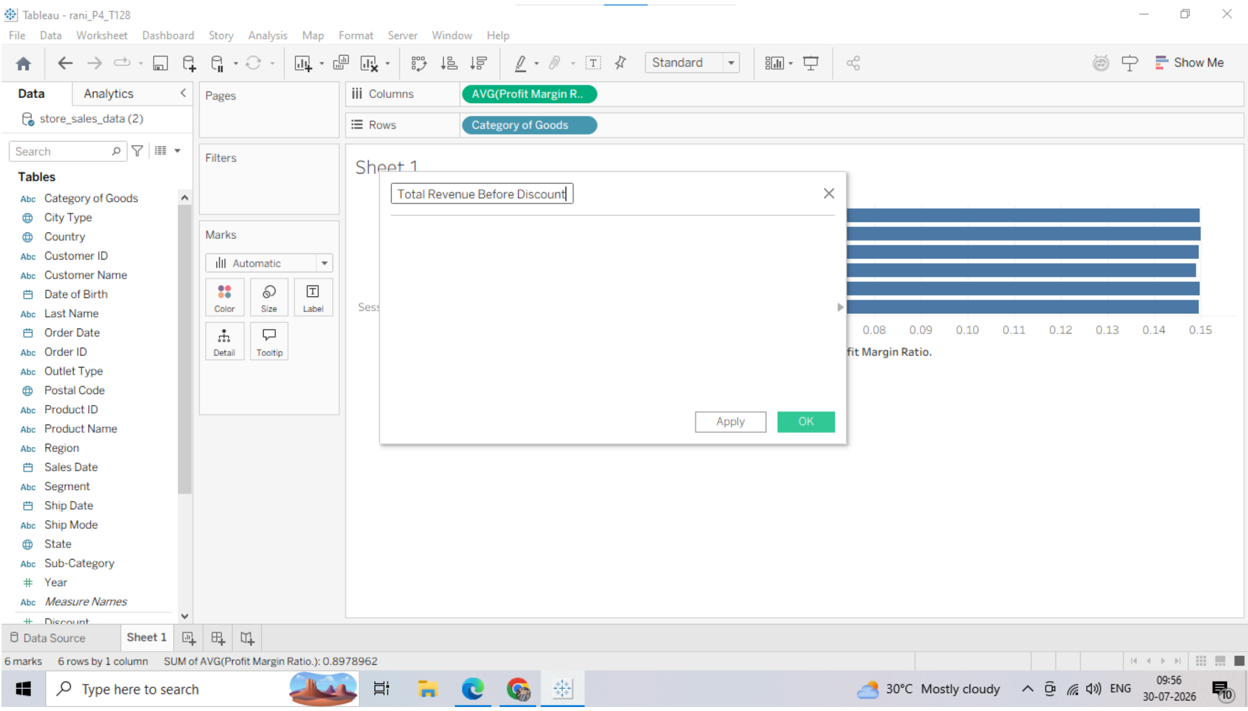
B: Performing Row-Level Calculations

1. Creating a Row-Level Calculated Field

1. Right-click anywhere in the blank white space of your Data Pane and choose Create Calculated Field...

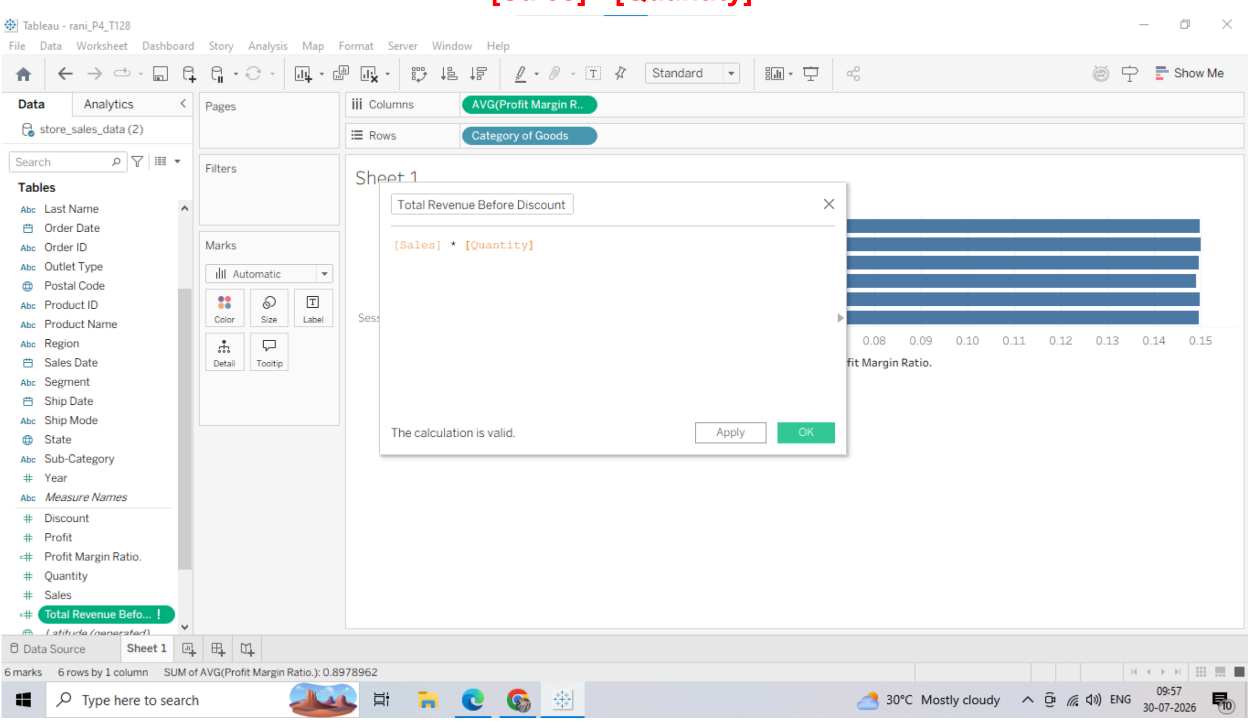


2. Name the field: Total Revenue Before Discount.



3. In the calculation formula canvas area, input the following expression:

[Sales] * [Quantity]



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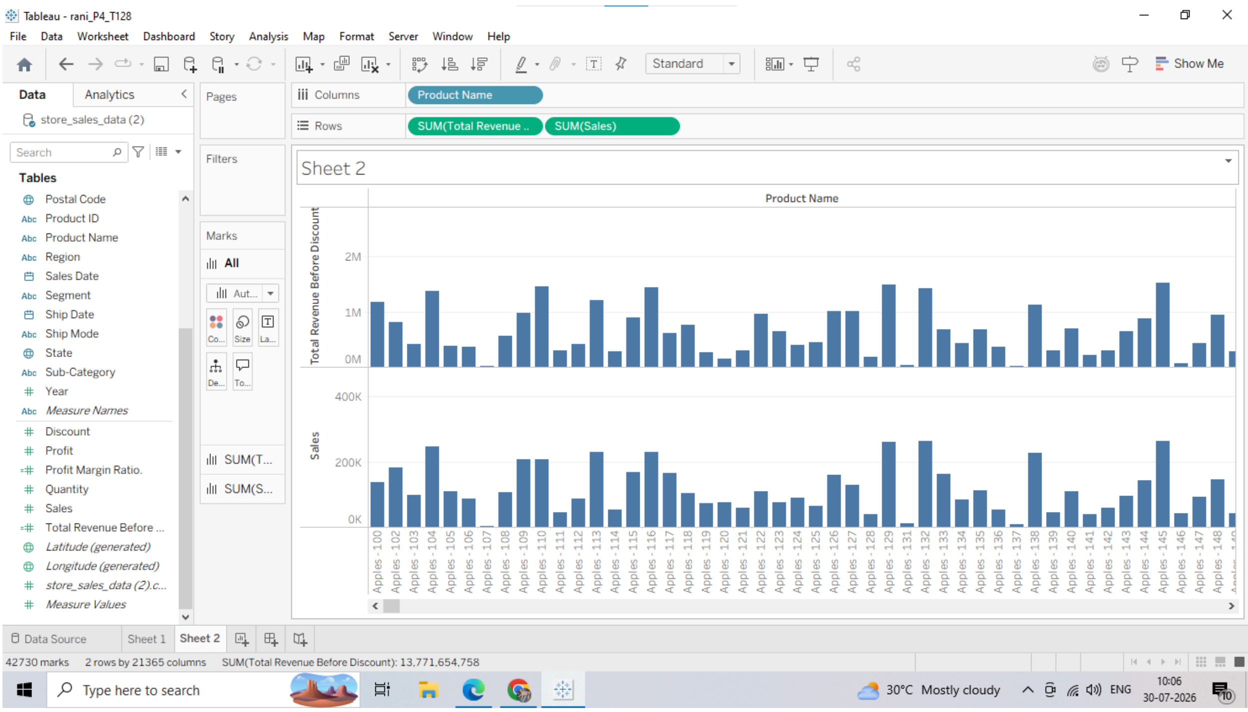
2. Putting the Field to Use

1. Drag the Product Name dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Total Revenue Before Discount calculated field and drop it onto the Columns shelf.
3. Drag the raw Sales measure pill and drop it directly onto the Columns shelf right next to the previous calculated pill to visually compare the two numbers.



Why we are doing it

A Row-Level Calculation instructs Tableau to perform a mathematical operation independently on every single record (row) inside the underlying store_sales_data.csv file before performing any visual sorting or overall data clustering. For instance, Tableau will go to row #1, multiply its specific sales number by its specific quantity, store that value, and repeat this check across all 10,000+ rows sequentially.

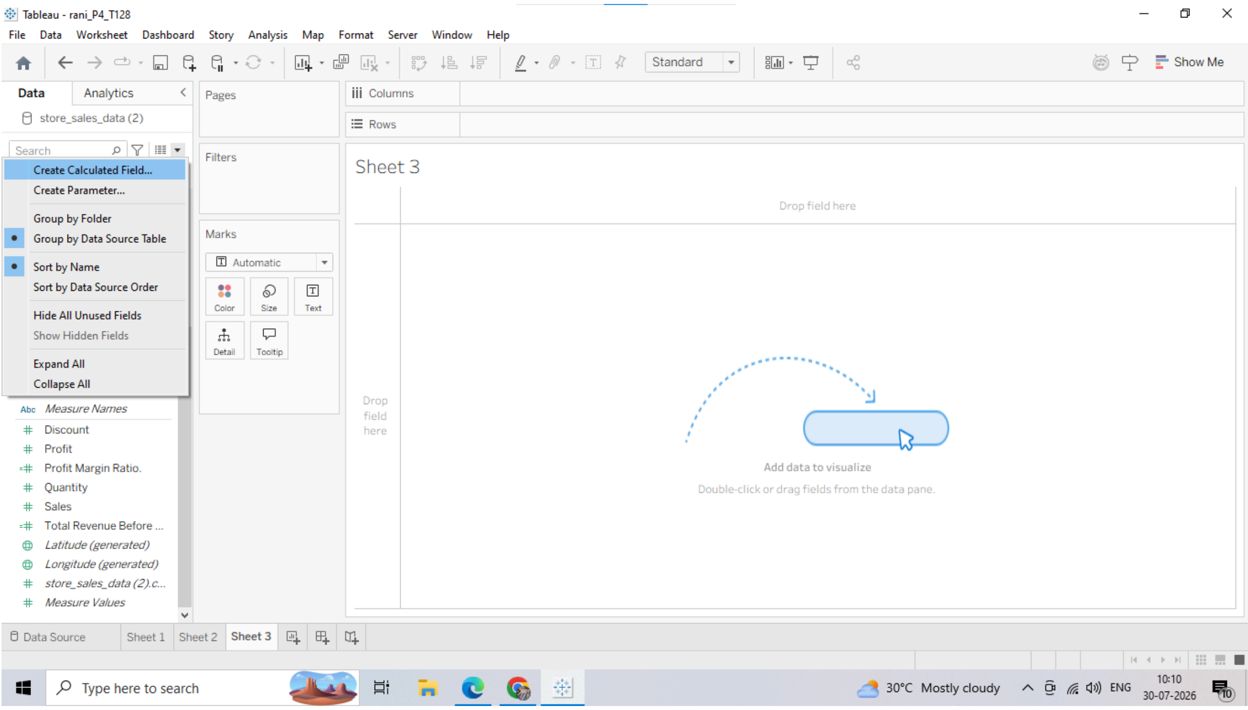
Usage

Use row-level calculations when you need to evaluate parameters that are strictly independent for every line item transaction (e.g., determining tax amounts per purchase order, calculating the total days elapsed between an **Order Date** and a Ship Date on each line, or multiplying base costs by transaction counts).

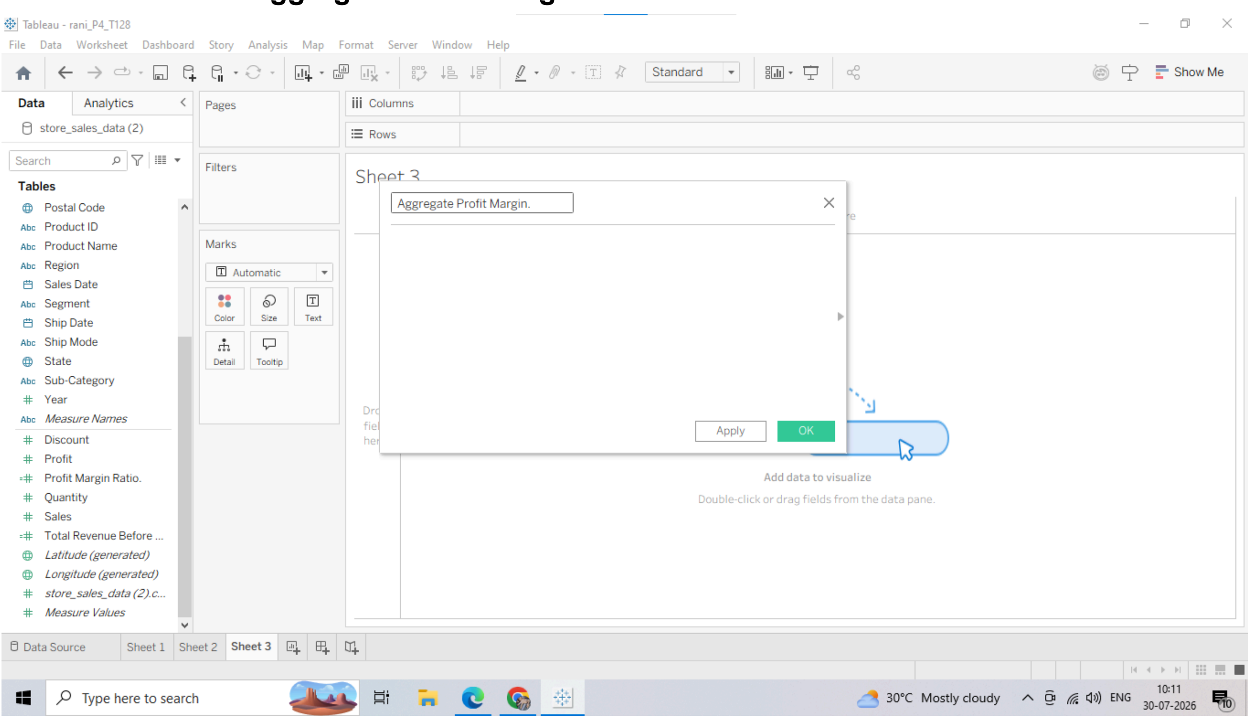
C: Performing Aggregate-Level Calculations

1. Creating an Aggregate Calculated Field

1. Right-click in the blank white space of your Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...



2. Name the field: Aggregate Profit Margin.



3. In the calculation formula window, type the following expression:
SUM([Profit]) / SUM([Sales])

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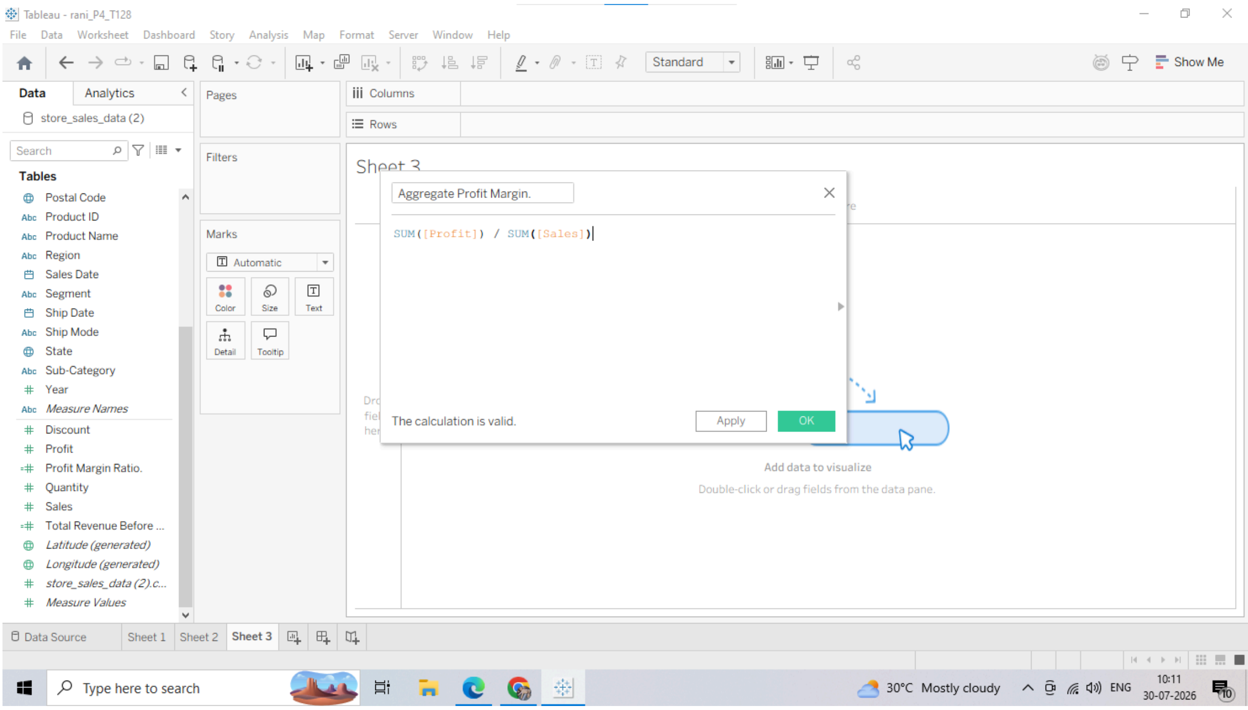
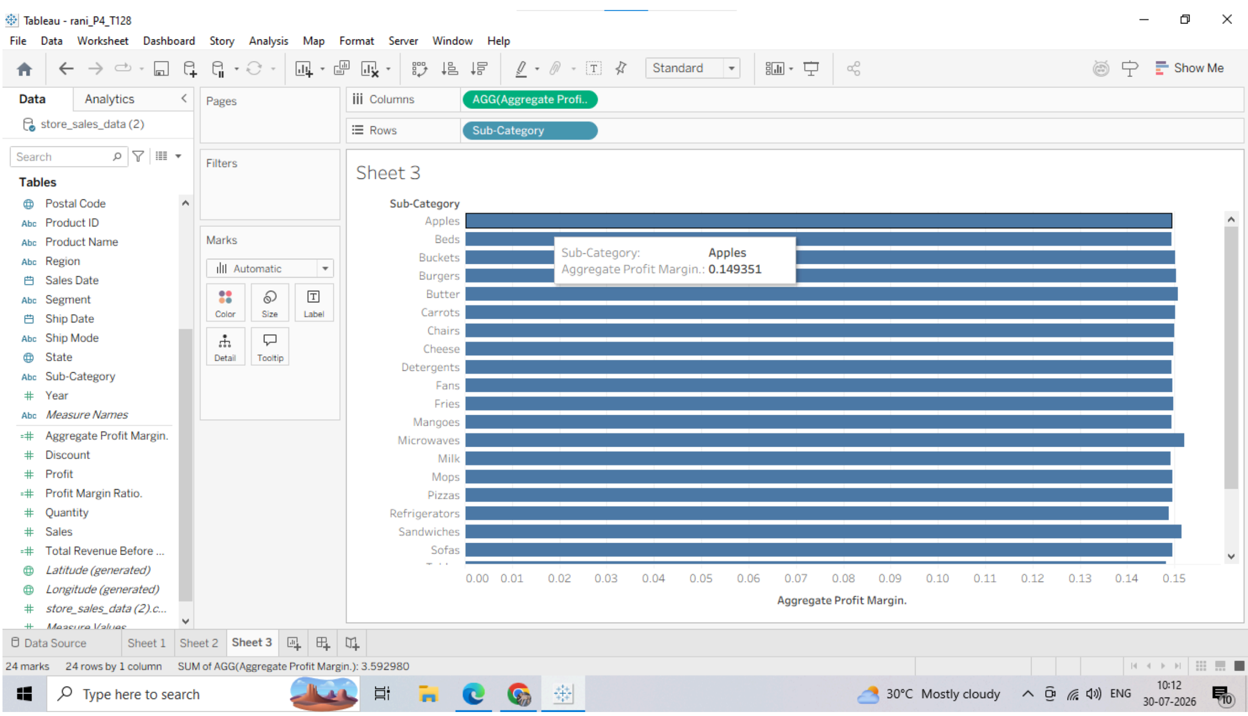


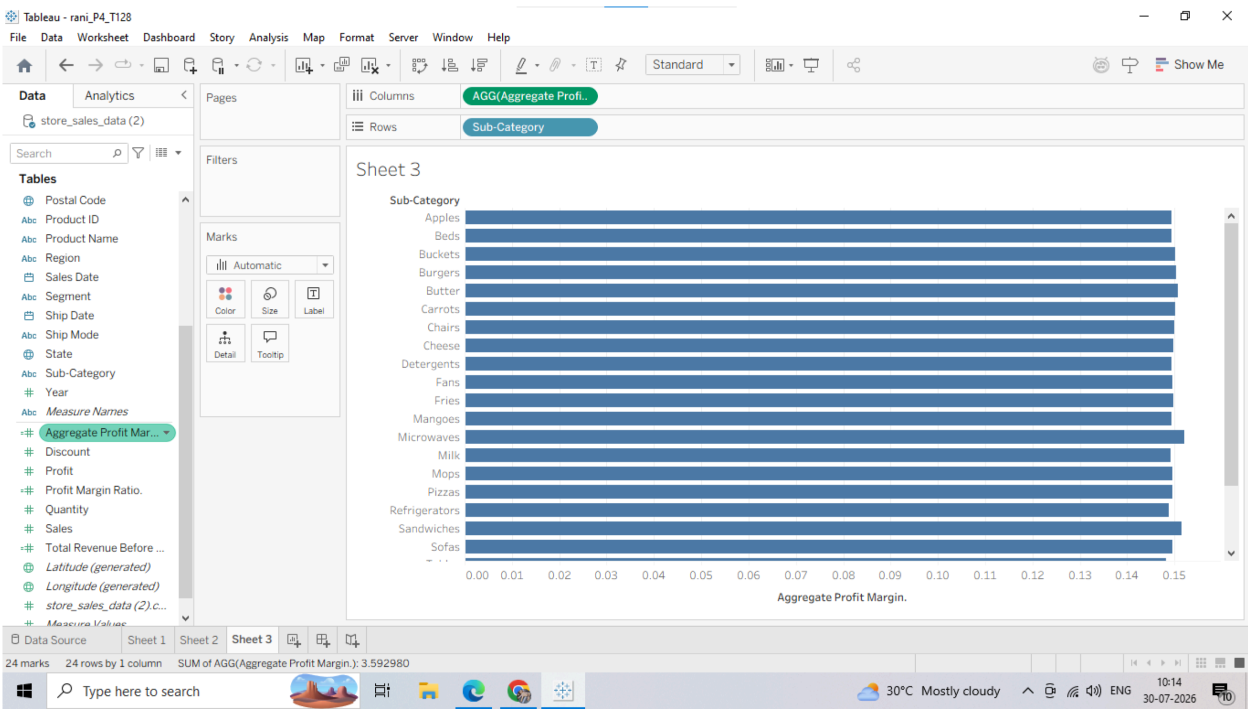
2. Putting the Field to Use

1. Drag the Sub-Category dimension from the Data Pane and drop it onto the Rows shelf.

2. Drag your newly created Aggregate Profit Margin calculated field and drop it onto the Columns shelf.



3. Observe closely: Look at the pill on the Columns shelf. It displays as AGG(Aggregate Profit Margin). Unlike standard fields, you cannot right-click this pill to change it to "Average" or "Count"—Tableau locks it as an aggregation because the math is already defined inside the formula.



Why we are doing it

An Aggregate-Level Calculation instructs Tableau to first sum up all the profits and all the sales for whatever category is displayed in your view, and then perform the division.
To see why this is critical, look at the difference between the row-level math from Step A and this aggregate step: Row-Level ([Profit] / [Sales]): Calculates the margin for each transaction individually ($10 / $100 = 10\%$). If you drop this onto a chart, Tableau takes the average of those percentages, which creates mathematical errors when orders are different sizes. Aggregate-Level (SUM([Profit]) / SUM([Sales])): Adds up the total profit of the category and divides it by the total sales of that category. This gives you the mathematically correct business profit margin.

Usage

Use aggregate calculations whenever you are creating ratios, percentages, or key performance indicators (KPIs) that must adapt accurately as you drill up or down into different visual categories.

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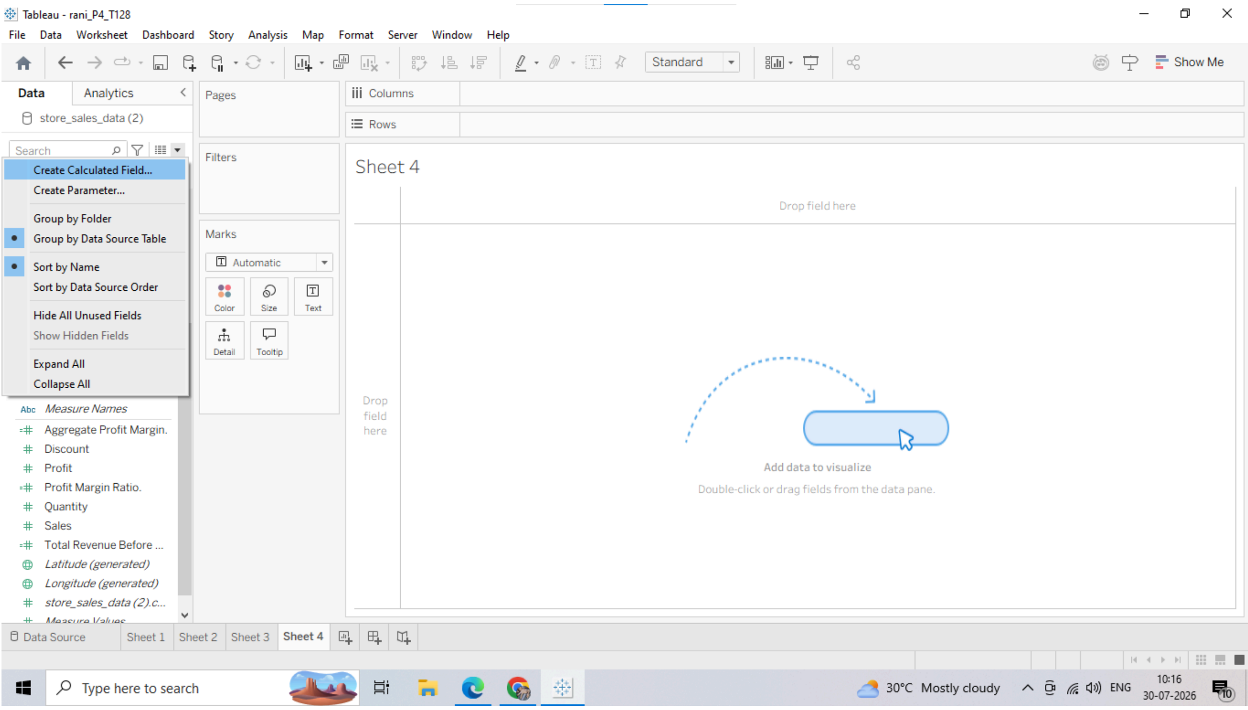
D: Implementing FIXED Level of Detail (LOD) Calculations

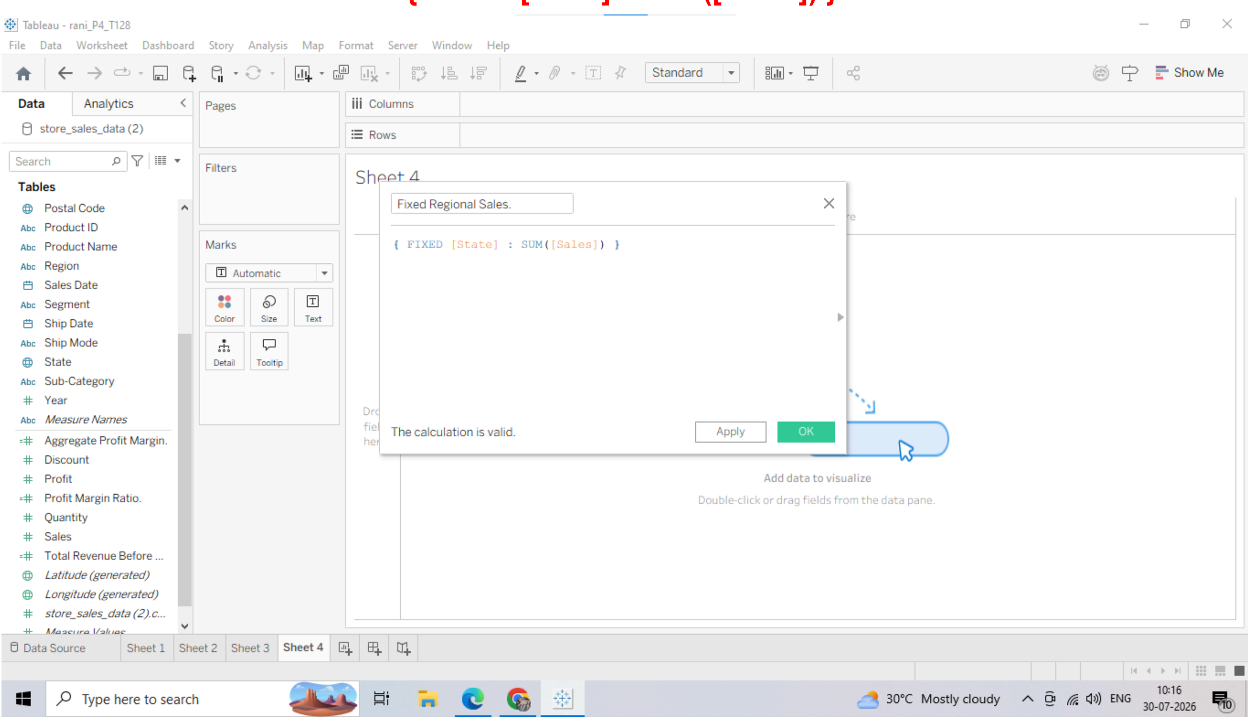
1. Creating a FIXED LOD Calculated Field

1. Right-click in the blank white space of your Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...



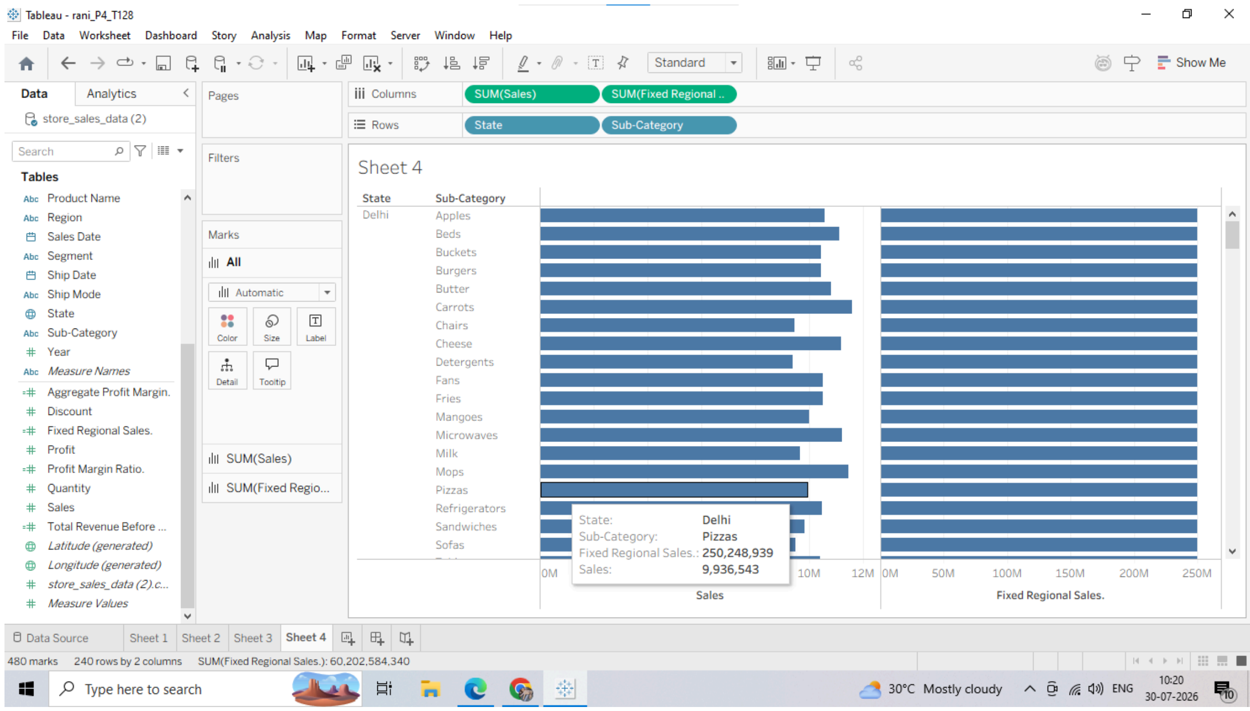
In the calculation formula window, type the following expression using curly braces {}:

{ FIXED [State] : SUM([Sales]) }



2. Putting the Field to Use

1. Drag the State dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag the Sub-Category dimension and drop it onto the Rows shelf, right next to State.
3. Drag the raw Sales measure pill onto the Columns shelf. (This shows the sales split down by each sub-category within that state).
4. Drag your newly created Fixed Regional Sales calculated field and drop it onto the Columns shelf, right next to the raw Sales pill.



Observe closely: While the raw Sales numbers change for every sub-category row, the Fixed Regional Sales number repeats the exact same value down the entire column for a given state. It completely ignores the Sub-Category breakdown in the view.

Why we are doing it

A FIXED Level of Detail (LOD) Expression computes a value using specified dimensions, without reference to the dimensions displayed in the worksheet view. By writing { FIXED [State] : SUM([Sales]) }, we tell Tableau: "No matter what rows or columns I add to my chart layout later, go straight to the data source, calculate the absolute sum of sales for each State, and lock that value down."

Usage

Use FIXED LOD calculations when you need to calculate market share percentages (e.g., dividing individual sub-category sales by the fixed total state sales) or when you want to establish a benchmark baseline value that must remain completely unchanged by filters or layout drills in your report.

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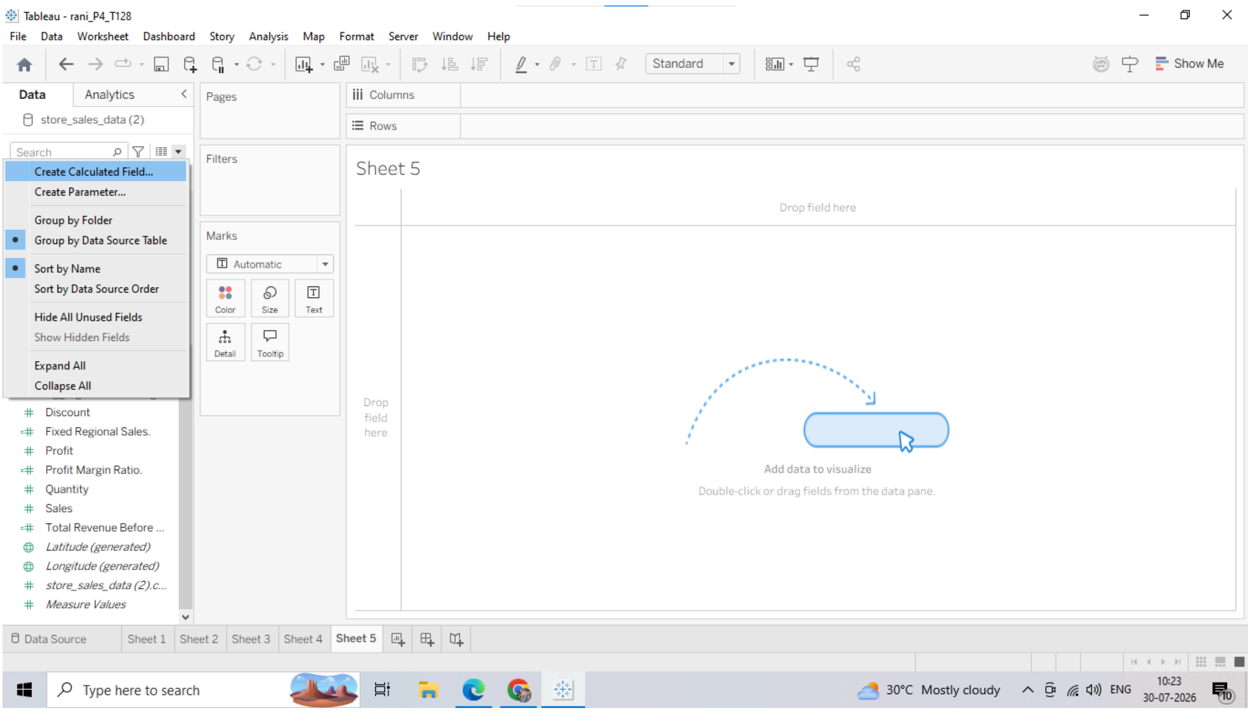
E: Using INCLUDE and EXCLUDE LOD Expressions

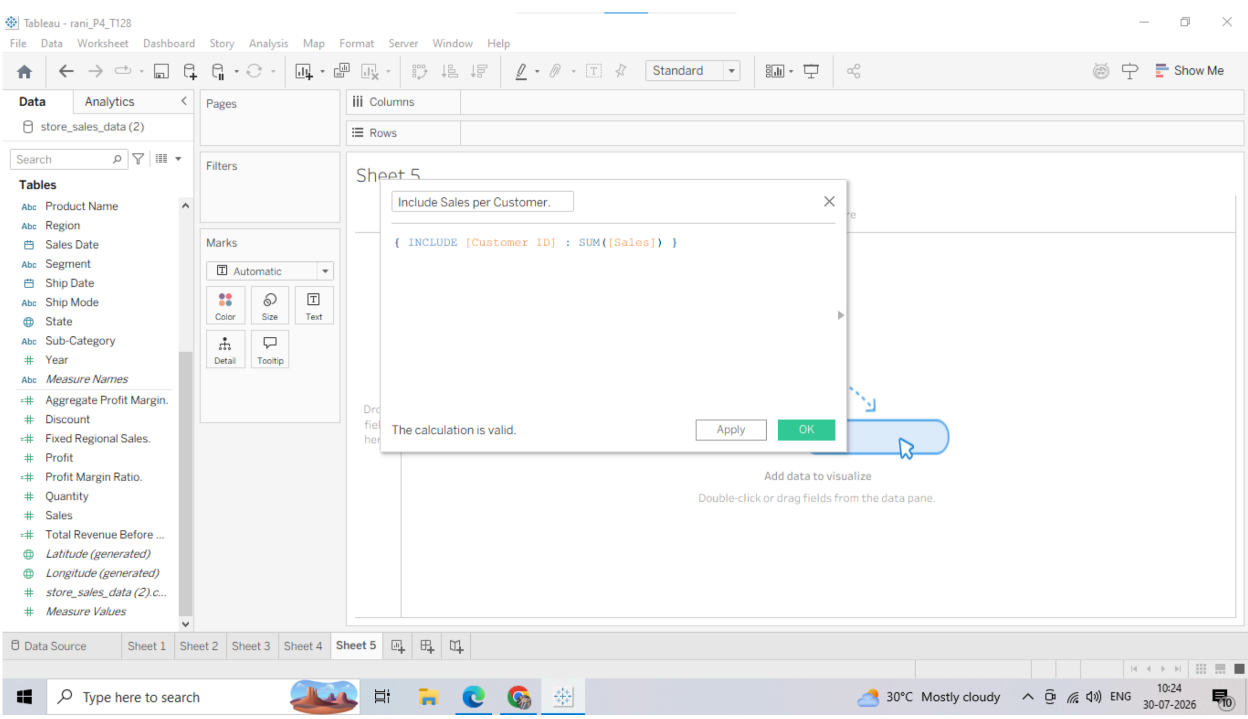
1. Creating an INCLUDE LOD Calculated Field

1. Right-click in the blank white space of your Data Pane and select Create Calculated Field...



In the calculation formula canvas area, input the following expression:

{ INCLUDE [Customer ID] : SUM([Sales]) }



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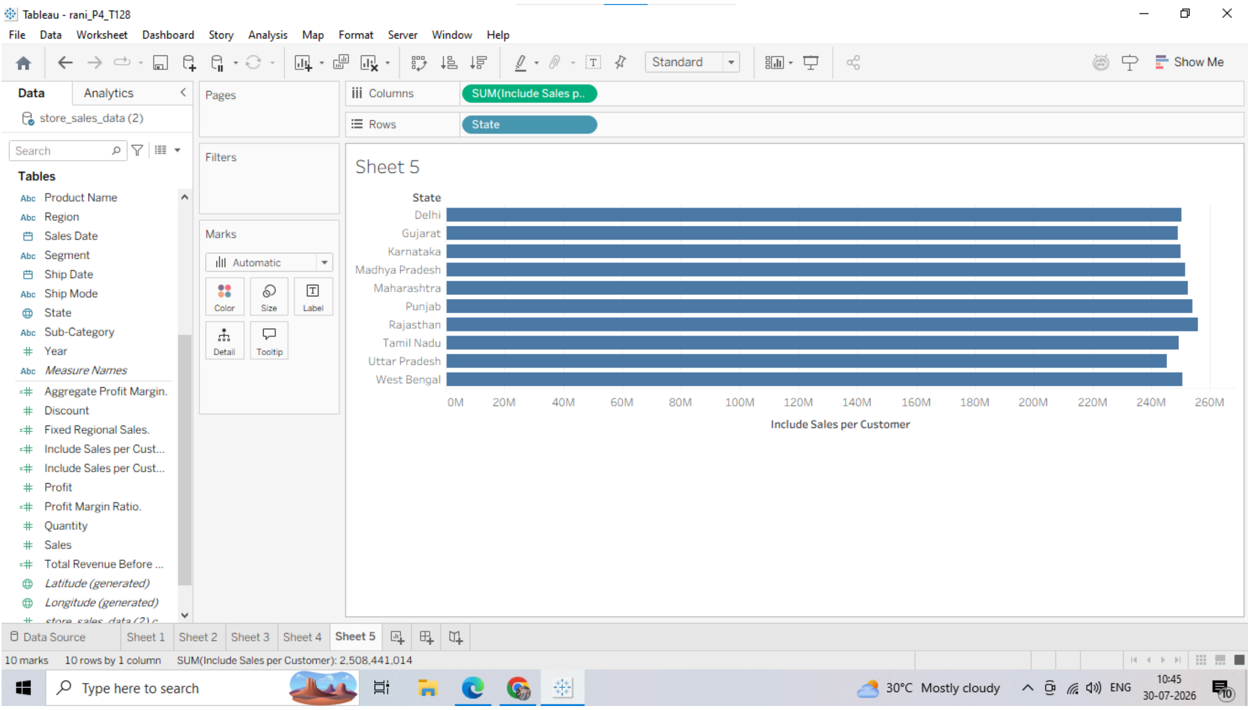
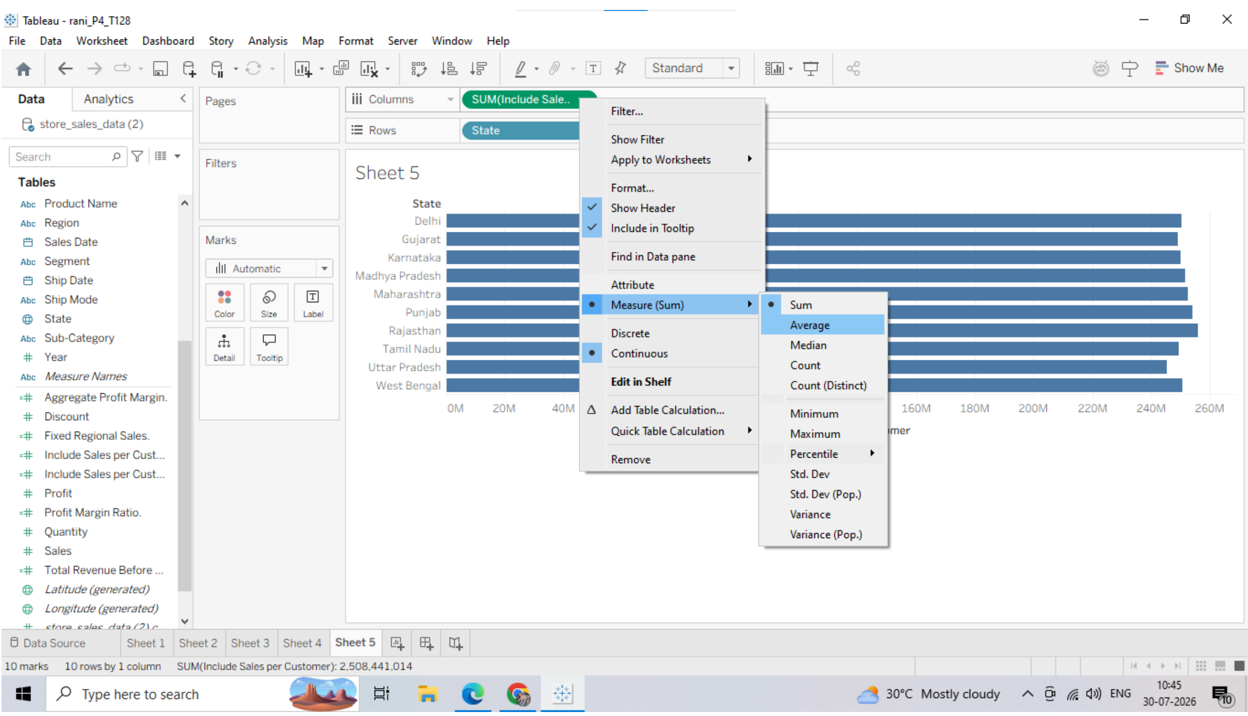
1. Drag the State dimension from the Data Pane and drop it onto the Rows shelf.
2. Drag your newly created Include Sales per Customer calculated field and drop it onto the Columns shelf.



Right-click the Include Sales per Customer pill on the Columns shelf, hover over Measure (Sum), and change it to Average.



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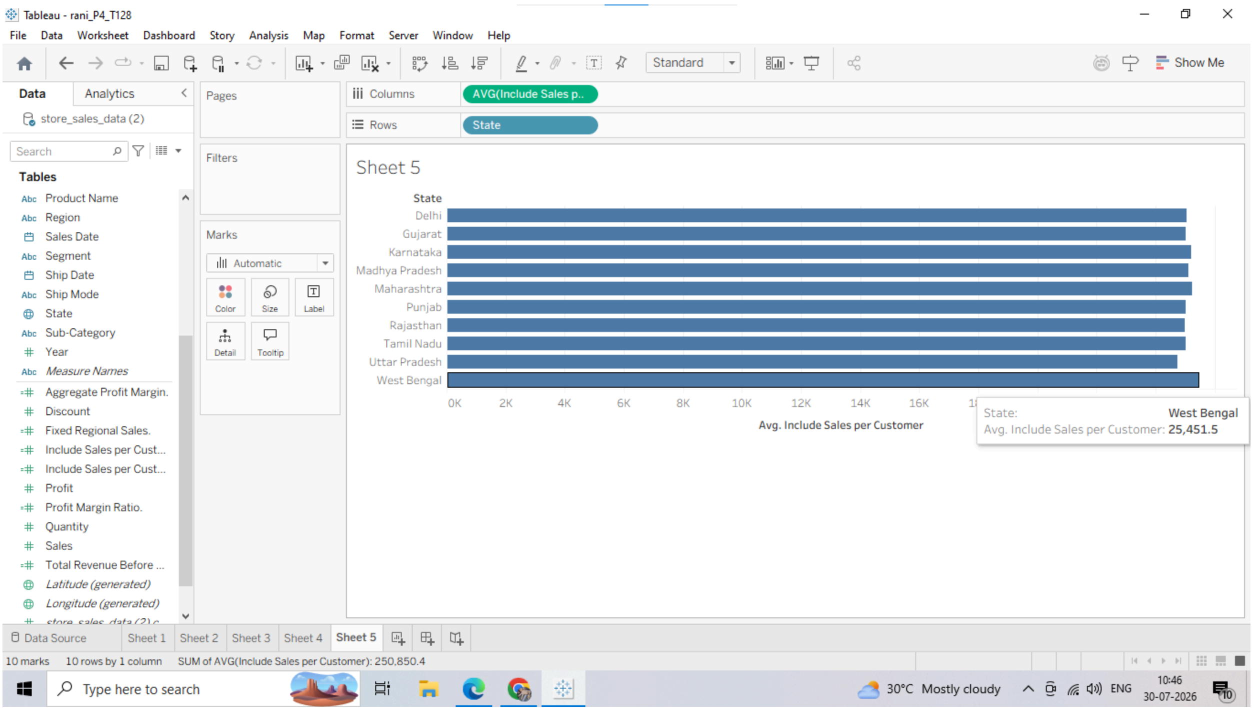

Why we are doing it: The INCLUDE LOD expression calculates a value using specified dimensions plus whatever dimensions are already present in the worksheet view. Even though Customer ID is not dragged onto our chart rows or columns, this expression forces Tableau to calculate the sum of sales for each individual customer first, and then average those customer totals out by each State. Usage: Use INCLUDE when you need to calculate an aggregate metric at a lower (more granular) level of detail than what is currently showing on your chart (e.g., calculating average transaction size per customer across broad regions).

3. Creating an EXCLUDE LOD Calculated Field

1. Open a new blank worksheet.
2. Drag Category of Goods onto the Rows shelf.
3. Drag Sub-Category onto the Rows shelf (right next to Category of Goods).

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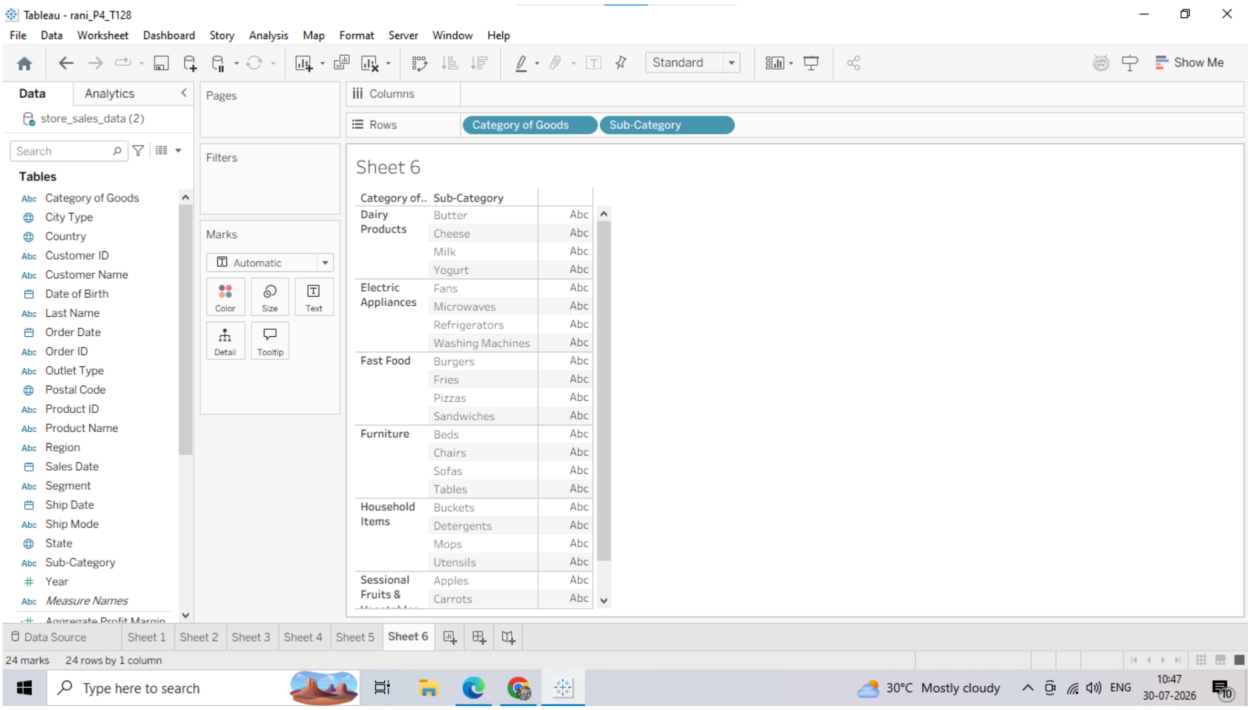
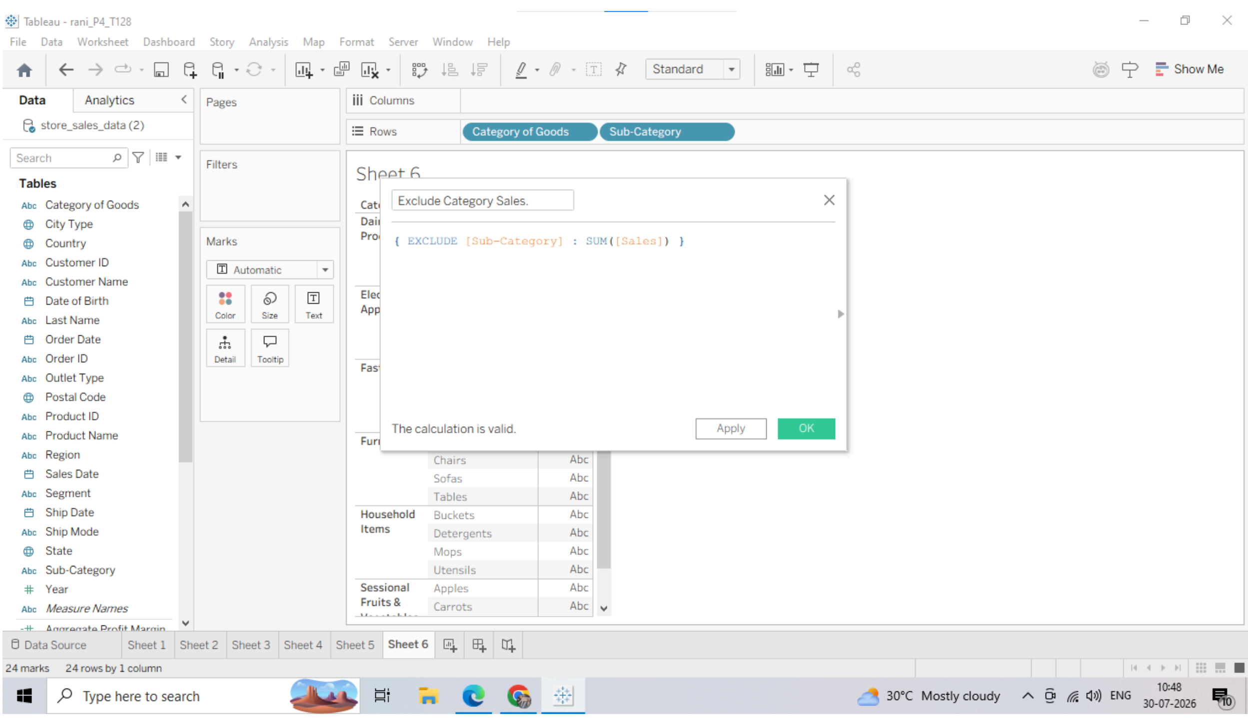


Right-click in the blank white space of the Data Pane or Click the Small Triangle Icon next to search bar and select Create Calculated Field...



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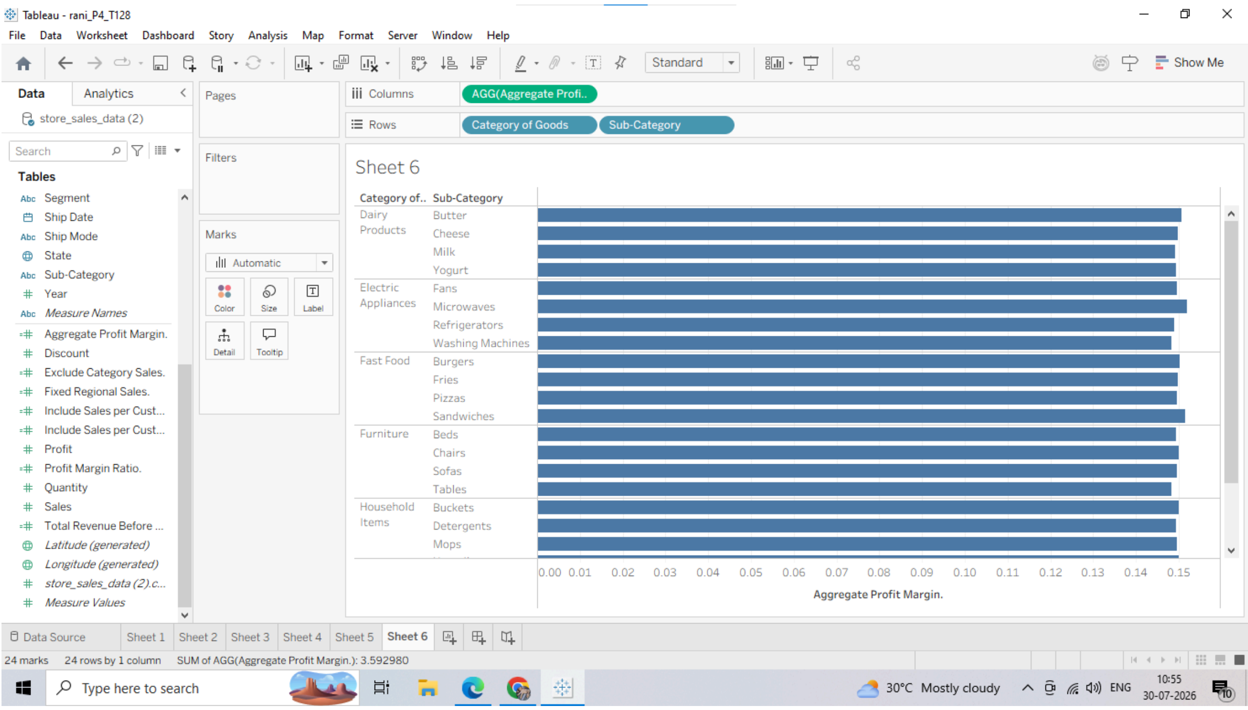


Why we are doing it: The EXCLUDE LOD expression instructs Tableau to intentionally ignore specific dimensions that are actively sitting on the worksheet rows or columns. By excluding Sub-Category, the values in this column will show the overall total sales for the main Category of Goods, remaining identical across all sub-category split rows.

Usage: Use EXCLUDE when you want to remove a highly specific dimension from an internal calculation to easily construct structural percentage comparisons (e.g., dividing a specific sub-category's performance against the broader parent category's total sales volume).

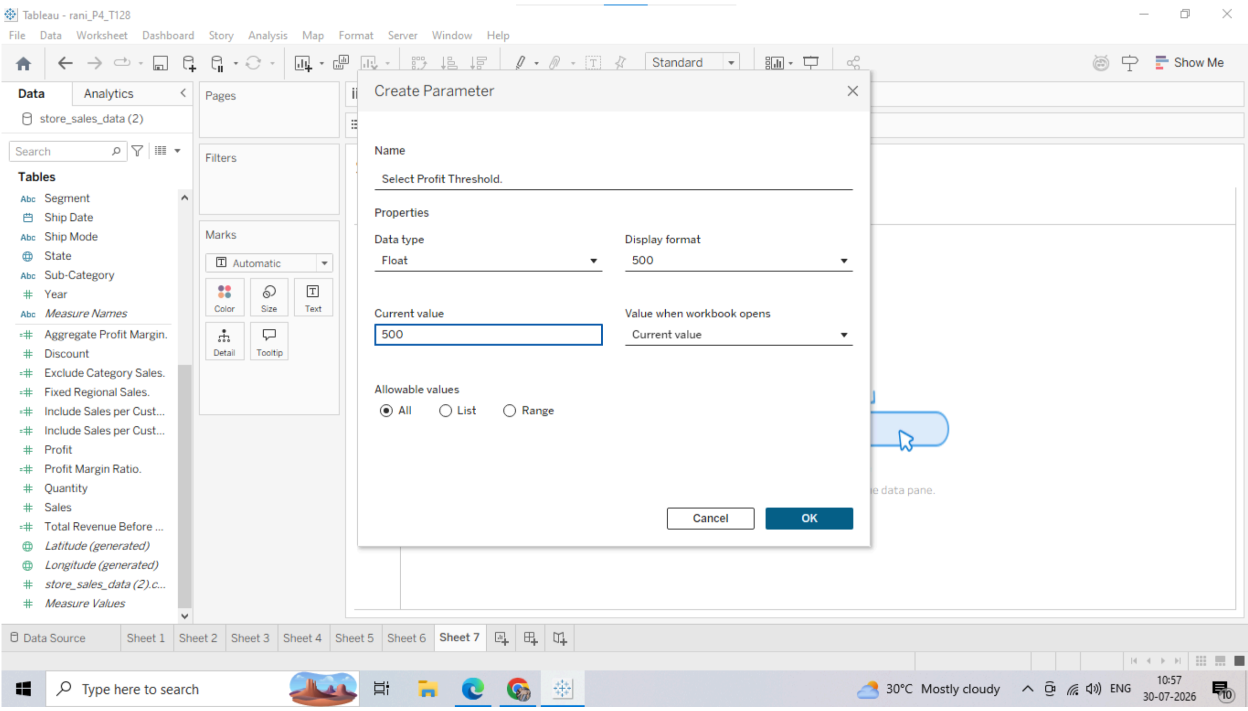
F: Creating and Using Parameters for Dynamic Analysis

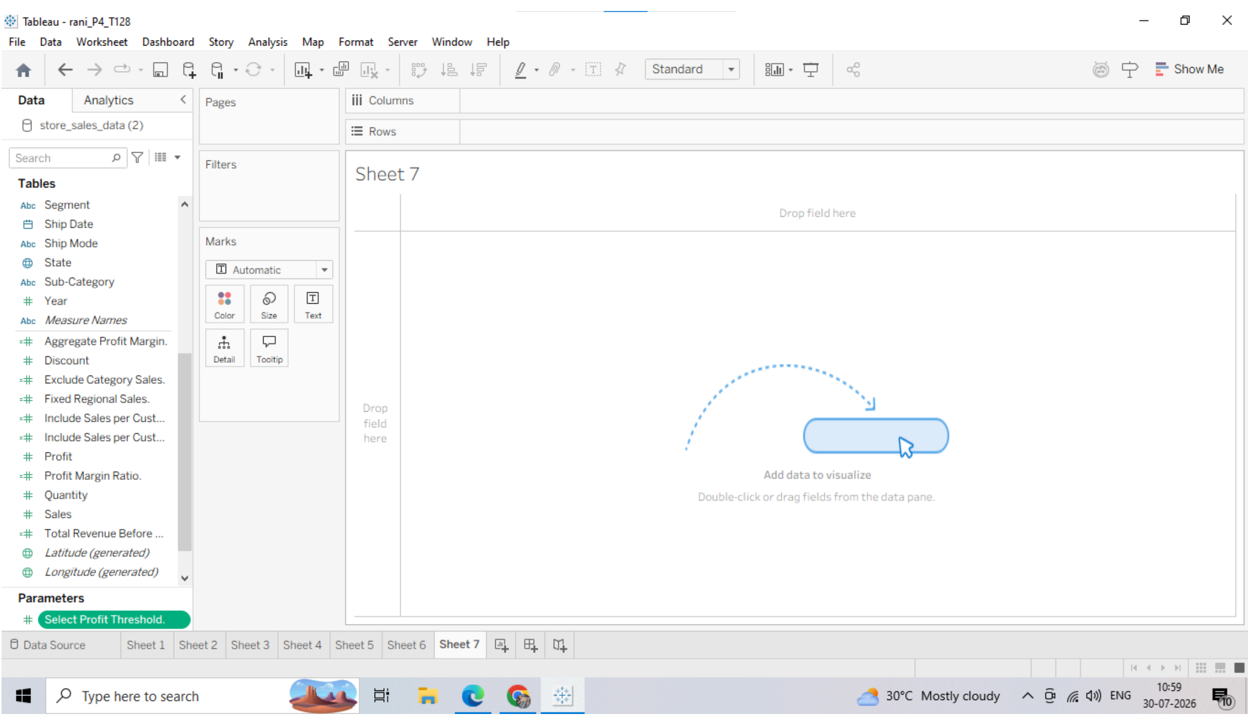
1. Creating a Parameter

Unlike calculated fields which evaluate data statically, a Parameter is a dynamic variable or workbook input control block (like a slider or dropdown) that allows users to alter calculations or values on the fly without changing the underlying architecture.

Let's build a parameter that lets users set their own dynamic Profit Target threshold:

1. Go to your Data Pane. Click the small drop-down arrow (small triangle) next to the search bar at the top and choose Create Parameter.





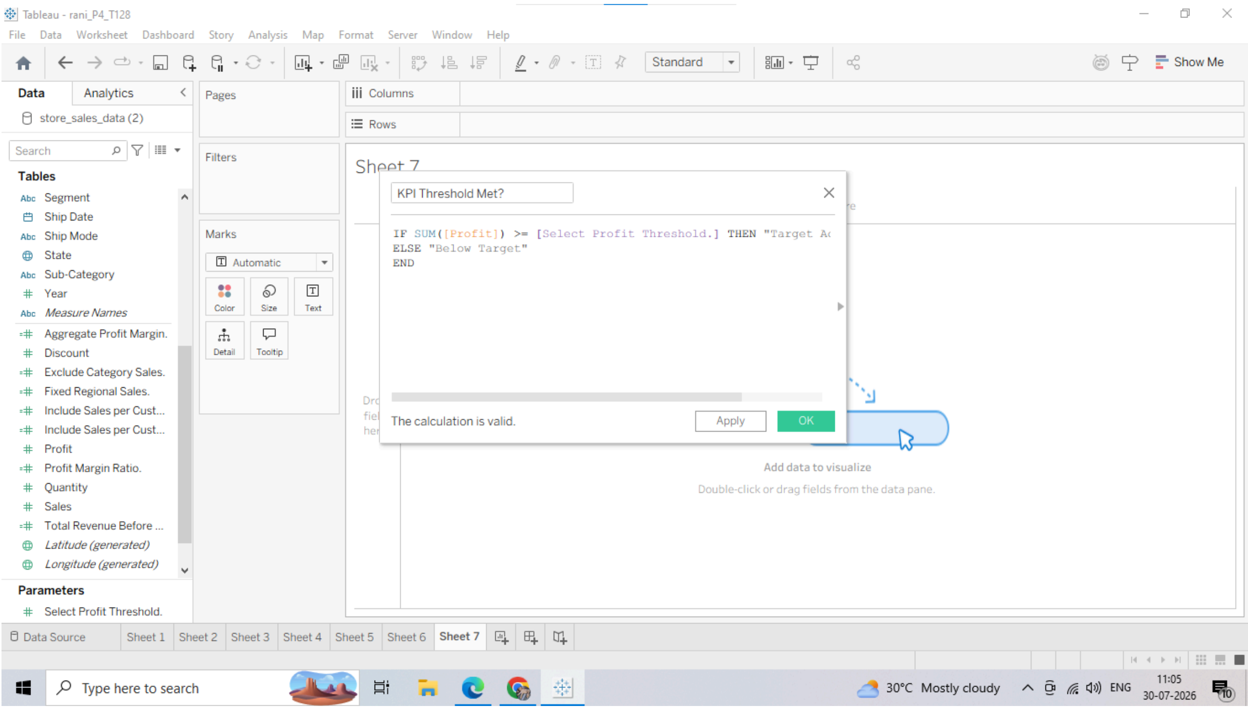
2. Linking the Parameter to a Calculated Field

A parameter does absolutely nothing until you actively tie it to a field layout or formula. Let's create a conditional formula that flags categories based on the parameter's current value:

1. Right-click in the blank white space of the Data Pane or Click on small triangle angle

next to search bar and select Create Calculated Field...

2. Name this field: KPI Threshold Met?.



4. Verify the formula is valid and click OK. 3. Putting the Interactive Parameter to Use

1. Open a new blank worksheet.
2. Drag the State dimension from your Data Pane and drop it onto the Rows shelf.
3. Drag the Profit measure and drop it onto the Columns shelf.
4. Drag your newly created calculated field KPI Threshold Met? and drop it directly onto the Color tile in the Marks card.

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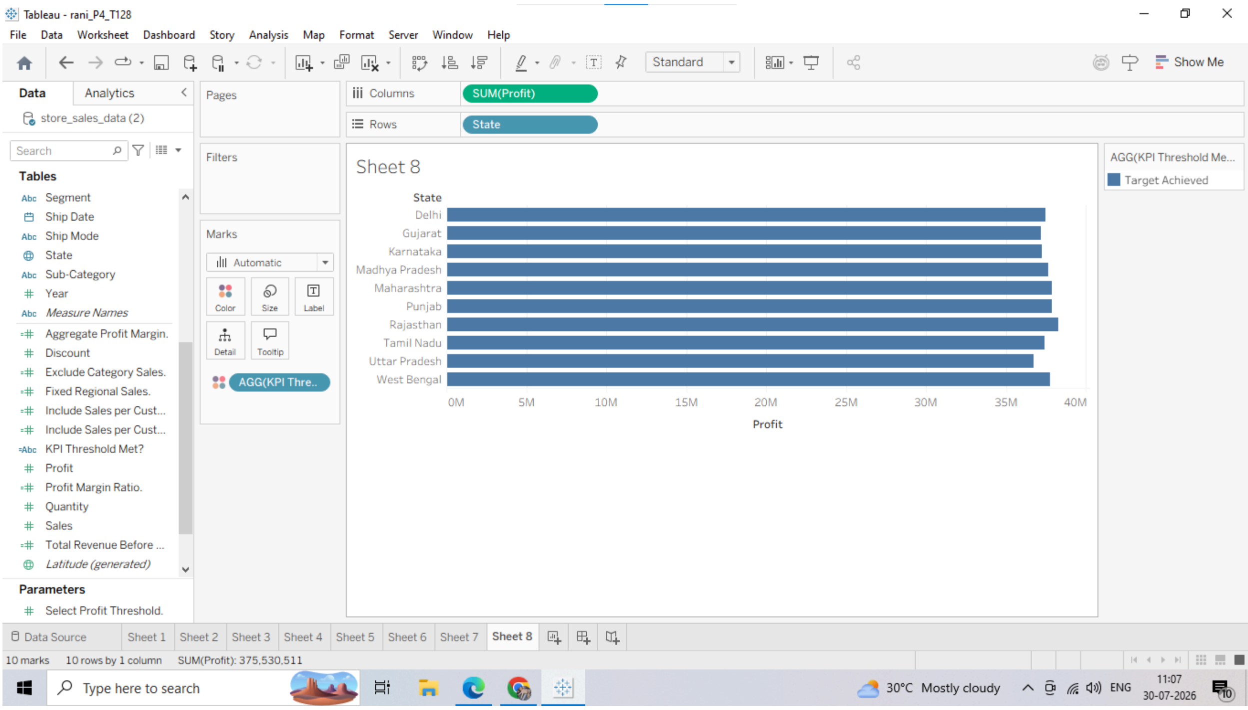
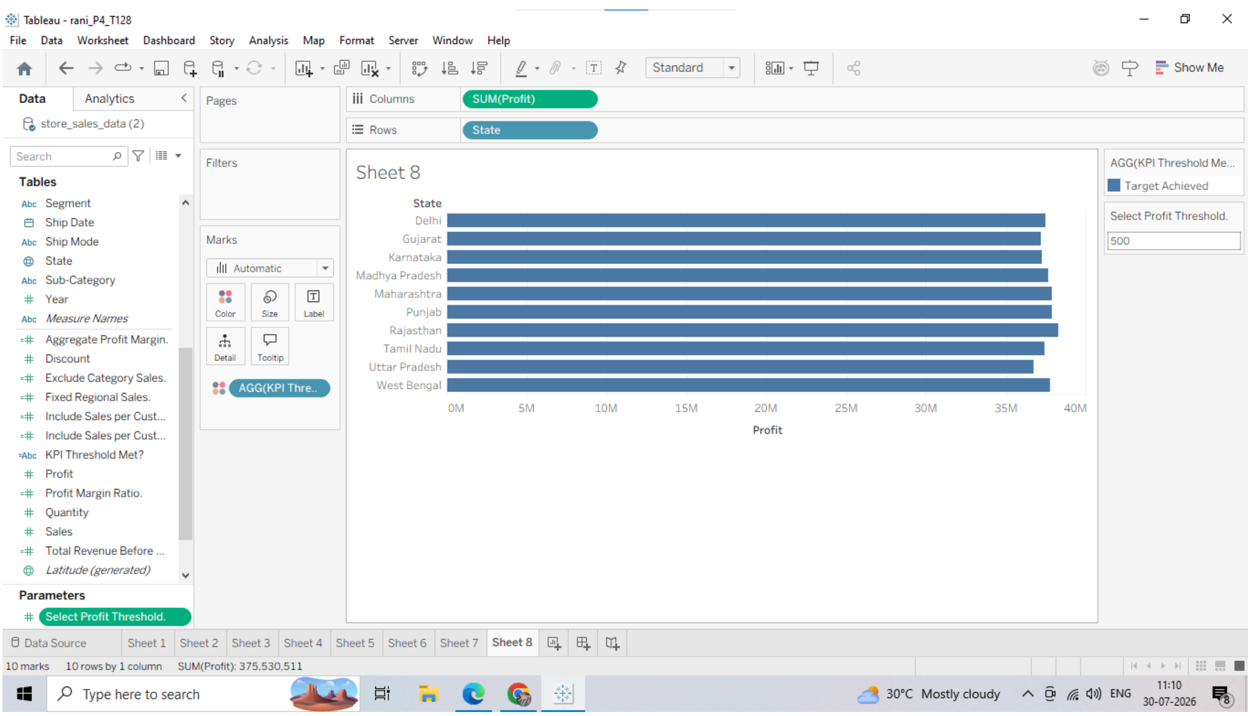


5. Show the Parameter Control: Go down to the very bottom of the Data Pane under the Parameters header section. Right-click on your Select Profit Threshold parameter block and choose Show Parameter Control.



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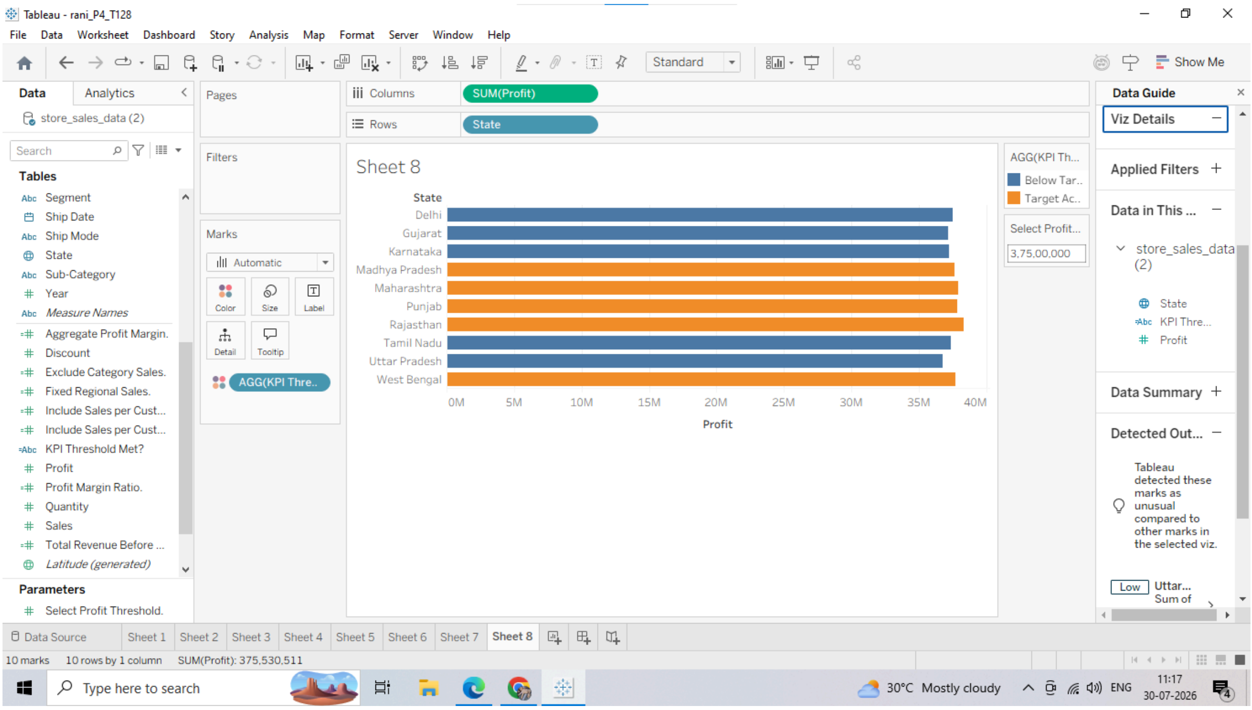
Observe closely: A text box slider will slide into the upper right corner of your sheet canvas view. Type a low value like 100 or a high value like 5000 into that parameter box and hit Enter **(NOTE: This is subjective to dataset in our case we typed 3,75,00,000 and the color changed instantly). The bars will instantly change color dynamically as states cross above or**
below the threshold you typed!



G: Apply Calculations to Fix Data Issues and Enhance Insights

1. Handling Null or Missing Values with IFNULL

In enterprise datasets, missing data or null entries can cause calculations to break or display
empty spaces in charts. We can use a logical calculation to substitute a clean value whenever
a null occurs.

Let's clean up potential missing indicators in our data:

1. Right-click in the blank white space of your Data Pane and select Create Calculated Field...

2. Name the field: Cleaned Discount. 3. Input the following expression:

IFNULL([Discount], 0)

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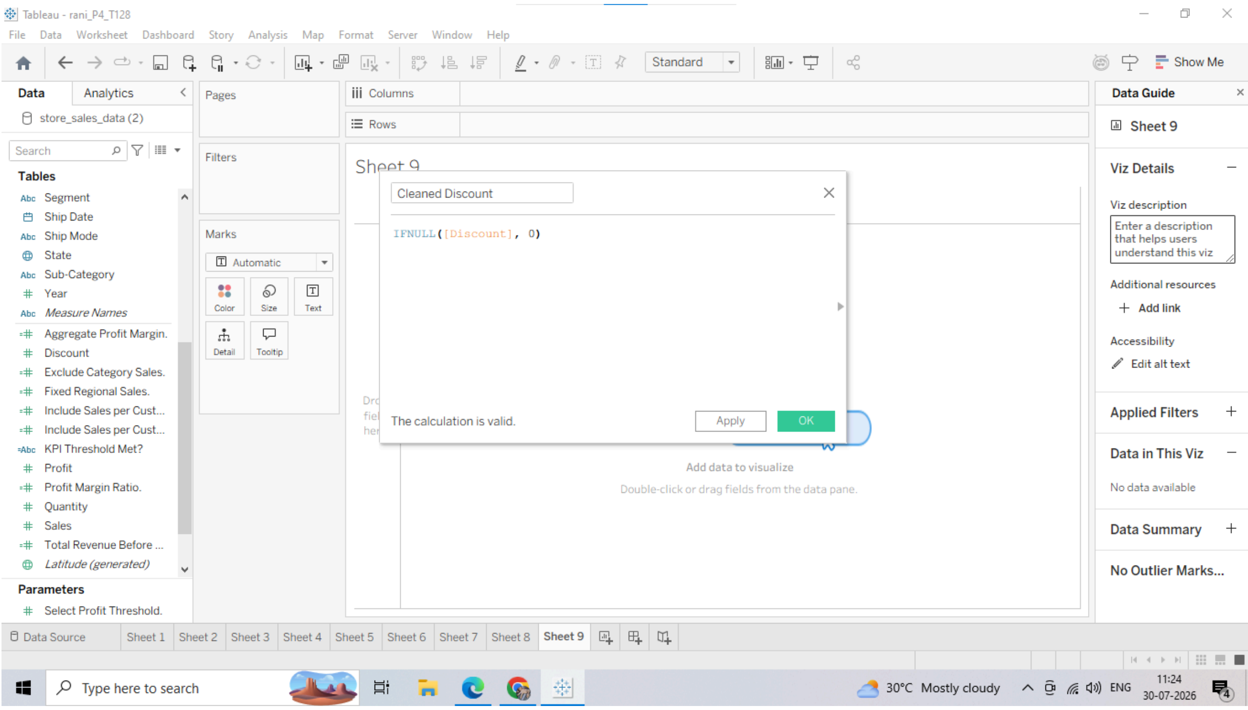
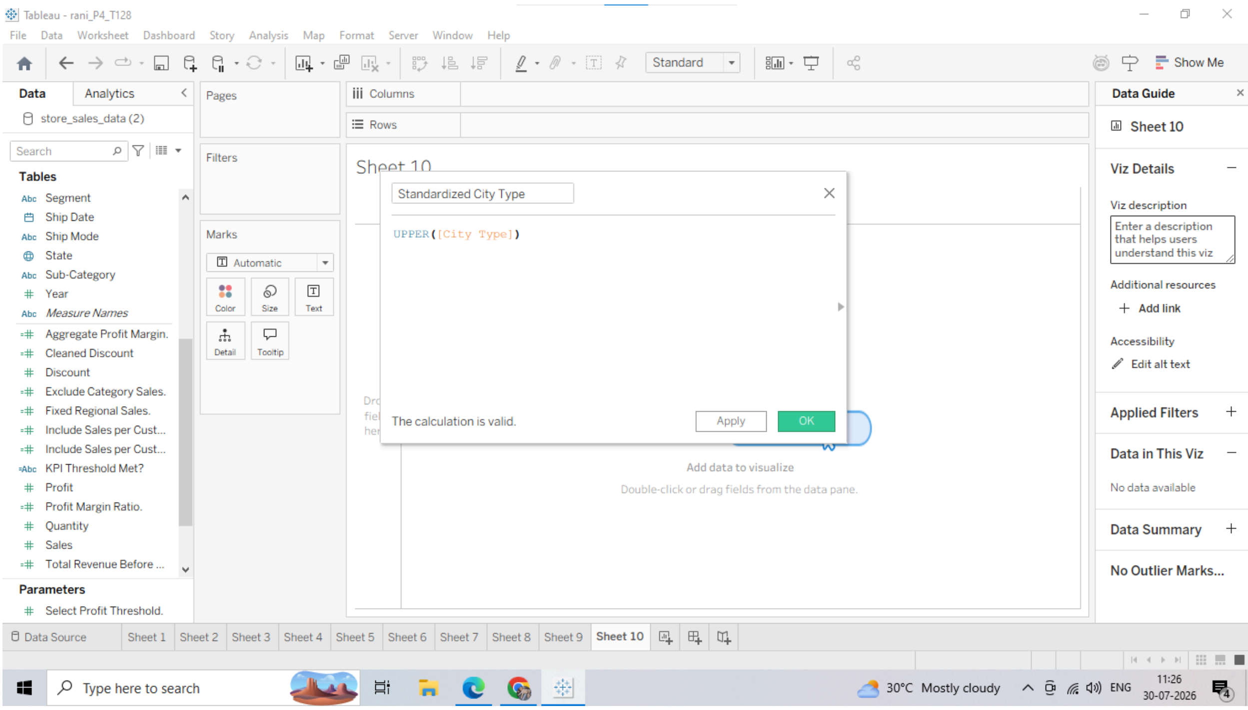


2. Segmenting Data for Insights using String Functions (UPPER/LOWER)
Sometimes database values are entered with inconsistent casing (e.g., "tier 1", "Tier 1", "TIER 1"), which causes Tableau to split them into separate visual blocks. We can force uniform compliance using a string calculation:
1. Create a new Calculated Field named Standardized City Type. 2. Input the following text transformation expression:
UPPER([City Type])



3. Putting the Cleaned Fields to Use

1. Open a new blank worksheet.
2. Drag your newly created Standardized City Type dimension from the Data Pane and drop it onto the Rows shelf.
3. Drag the Cleaned Discount calculated field onto the Columns shelf. Change its aggregation from Sum to Average by right-clicking the pill.

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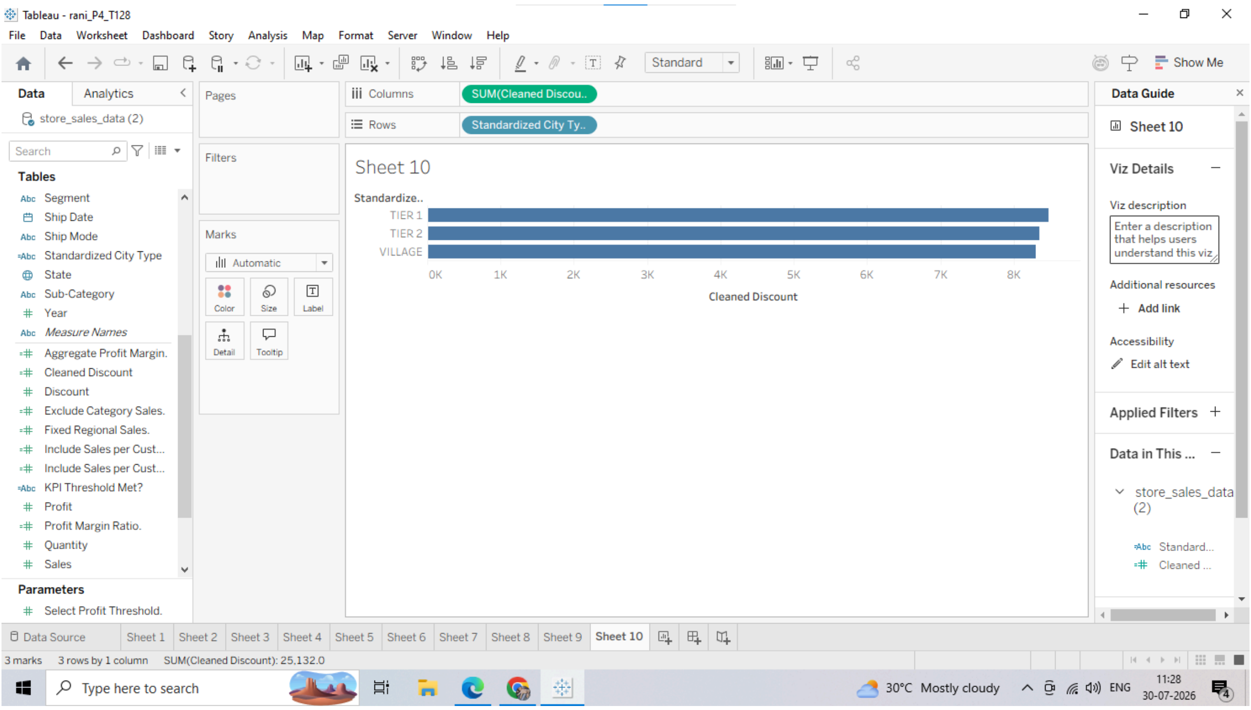

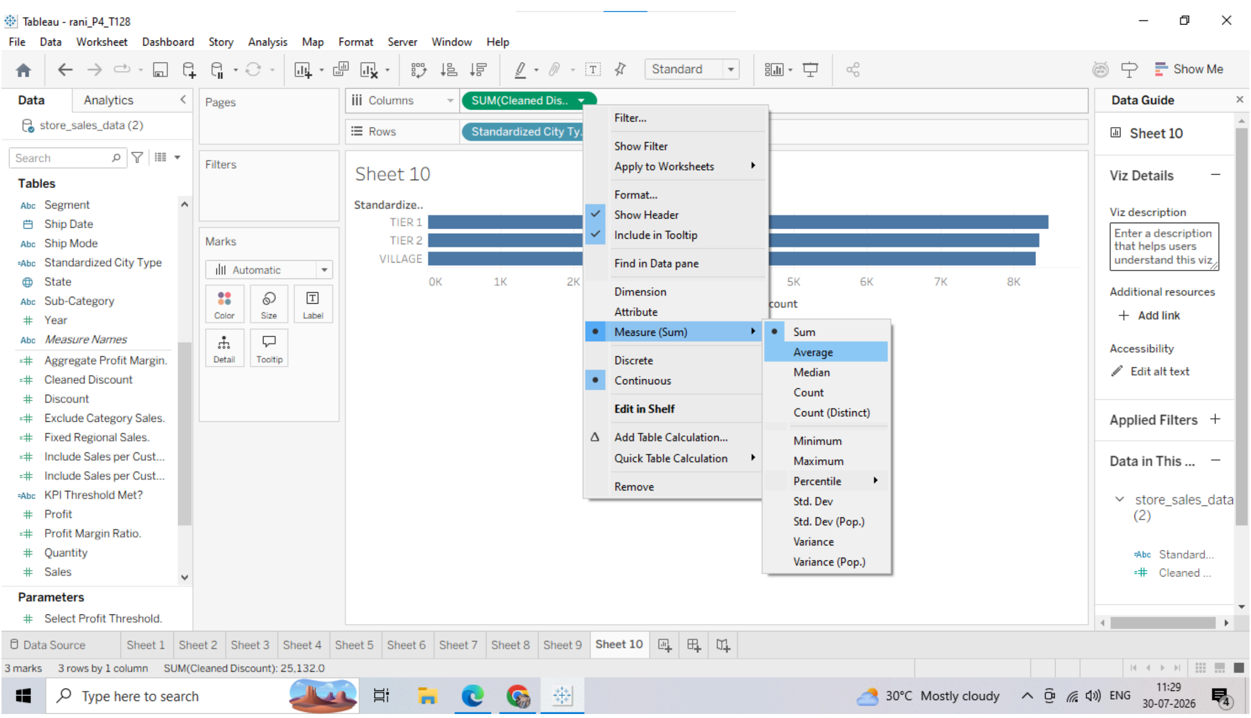

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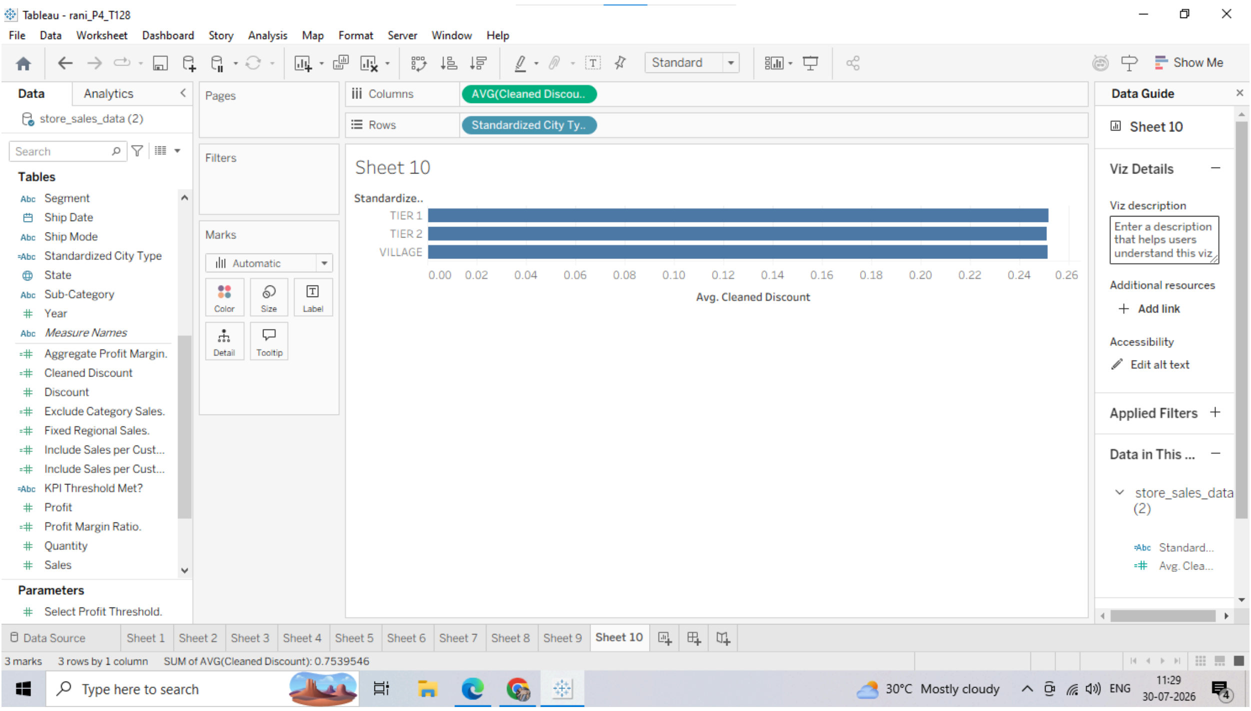

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